CPTG SkyPi: Decoding the CMB with Geometry

From Fixed Geometric CMB Solution to High-Resolution Planck Map and Spectrum Closure

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Abstract

The cosmic microwave background (CMB) is the relic radiation of the early universe. Small temperature variations across the sky form an angular-power spectrum whose acoustic peaks, relative amplitudes, and small-scale damping provide a stringent test of cosmological structure. Standard ΛCDM analyses calculate this spectrum from a cosmological model and estimate its parameters by comparing theoretical spectra with observations. Curvature Polarization Transport Gravity (CPTG) offers a fixed geometric alternative for the comparison: ordinary matter and radiation remain physical sources, while curvature polarization and transport provide additional geometric response. Its cosmological construction begins with one native Pi branch and registered acoustic and CMB comparison coordinates, rather than an adjustable cosmological row for each Planck analysis. *CPTG SkyPi* realizes this construction in a local workbench for temperature-map and angular-power analysis. It holds the registered source fixed while applying the selected Planck beam, mask, pixel, and harmonic operators; it evaluates spectral agreement, reconstructed temperature distributions, and map residuals. The map reconstruction retains the observed harmonic phases as spatial coordinates, while the fixed CMB power envelope determines the corresponding comparison amplitudes. The observational program spans Commander PR3/PR4, SEVEM, SMICA, SMICA-NOSZ and NILC, together with foreground-subtracted frequency products, HFI measurements and Galactic-dust observations. Four Planck PR3 component-separated odd–even analyses at NSIDE 2048 and $\ell_{\max} = 4096$, using the same common-intensity/OE mask and apodization protocol, yield map correlations of 0.993454–0.995103 and retained-region residual root-mean-square values of 8.8105–10.2111 μK; additional products are tested under their separately declared operators. A same-source Commander PR4 resolution series from NSIDE 256 through 4096 reduces the absolute mean-power difference over $450 \leq \ell \leq 511$ from approximately 2.57% to 1.10% before stabilizing. Because the geometric comparison coordinates remain fixed, new map products and resolutions do not entail a new cosmological parameter search or native CPTG perturbation calculation. SkyPi makes these geometric CMB comparisons reproducible directly from the selected Planck observations.

1 Introduction

1.1 The observable: a microwave sky and its power envelope

The cosmic microwave background (CMB) is radiation released when the early universe became transparent, now observed as a nearly uniform microwave sky with a mean temperature of about 2.725 K. Its small temperature fluctuations, typically expressed in microkelvin (μK), preserve information about the physical conditions and angular structure of the early universe. The European Space Agency's *Planck* mission measured this radiation over the full sky in multiple frequency bands. To study the primordial temperature fluctuations, the measured maps must be separated from Galactic emission, instrumental effects, and other contributions. Commander, SEVEM, SMICA, and NILC are distinct Planck component-separation products used for this purpose [6, 7].

A temperature map shows where relatively hot and cold regions appear. A different observable, the *angular-power spectrum*, describes how much temperature-fluctuation power occurs at each angular scale. On a sphere, the temperature field is decomposed into spherical harmonics, indexed by multipole ℓ; low ℓ describes broad angular patterns and high ℓ describes finer ones. The quantity C_ℓ^{TT} measures mean temperature power at a given multipole. It is often displayed as $D_\ell = \ell(\ell+1)C_\ell^{TT}/(2\pi)$, in units of $\mu\mathrm{K}^2$. In this paper, the *CMB power envelope* means the scale-dependent structure of this temperature-power distribution. It is not a map of the individual hot and cold spots, nor does it fix their positions on the sky.

The envelope is far from featureless. The succession of acoustic peaks and troughs records oscillatory behavior of the early photon–baryon medium; their angular positions, relative heights, and eventual small-scale damping constrain how a cosmological construction connects early-universe physics to the sky we observe. An acceptable comparison must therefore account for power *across angular scales*, not merely yield a plausible-looking temperature map. It must also distinguish the celestial signal from the effects of the detector response, foreground processing, sky mask, and noise.

1.2 The cosmological problem and the CPTG geometric response

In the standard ΛCDM framework, a small set of cosmological parameters describes the background and initial perturbations; numerical evolution determines the resulting theoretical CMB spectra, and observational likelihoods infer the supported parameter values. The six-parameter baseline describes the Planck temperature data well when the appropriate experimental, foreground, and statistical treatments are included. Its power-envelope agreement is thus obtained within a *parameter-inference framework*. The distinct question addressed here is whether a *registered geometric construction*, retained unchanged across selected Planck products and analysis resolutions, can exhibit comparable angular-power and temperature-field correspondence without re-estimating its geometric coordinates for each test.

Curvature Polarization Transport Gravity (CPTG) is a scale-structured gravitational framework in which ordinary matter, radiation, pressure, and other physical stress–energy remain the material sources. CPTG assigns additional gravitational response to the geometry itself through *curvature polarization* and *curvature transport*: the former describes the reorganization of sourced curvature, and the latter the retention or propagation of that organized response. Its parent gravitational description retains the Einstein term, with General Relativity recovered continuously when the additional geometric response becomes negligible [4]. CPTG is therefore proposed as a geometric account of effects conventionally described using additional cosmological or gravitating components, while retaining the ordinary physical sources.

In cosmology, the CPTG construction is organized around *one native Pi-based geometric branch*. Acoustic, luminosity, standard-ruler, and CMB descriptions are specified projections or comparison coordinates of that branch, not separately fitted cosmologies [1]. Two of its dimensionless coefficients and the native normalized expansion coordinate are

$$B_\pi = 1 - \frac{1}{\pi}, \qquad T_\pi = \frac{1}{\pi}, \qquad E_\pi^2(z) = B_\pi + T_\pi(1+z)^\pi, \tag{1}$$

where z denotes redshift. The acoustic comparison uses the registered exponent $p_{\mathrm{ac}} = 3 - \pi/100$ in place of the native exponent π. These relations illustrate the geometric starting point; the full chain of CMB-facing comparison coordinates, numerical source inputs, and their provenance is defined in Section 3. In particular, the numerical CMB power envelope is constructed through the registered comparison procedure, rather than being identified with the expansion equation alone.

1.3 From fixed geometry to the measured CMB

SkyPi is the computational realization of this program. The *source side* supplies the registered CPTG temperature-power envelope and holds its geometric comparison coordinates fixed. The *measurement side* supplies an actual Planck temperature map together with the relevant beam response, sky mask, pixel resolution, and harmonic limits. By applying these operators consistently, SkyPi can compare the fixed geometric envelope with the angular power measured in the selected observation. A change from one Planck product to another changes the observational treatment, not the registered CPTG geometry.

For map-space inspection, the measured harmonic phases supply the spatial coordinates of the observed hot and cold regions; the fixed SkyPi envelope supplies the comparison amplitudes. Consequently, strong map correlation must be interpreted together with the power-spectrum comparison, temperature-statistics distributions, and signed residuals. The observed-phase reconstruction is not a claim that Pi constants alone specify the celestial positions of individual temperature spots. Complementary half-mission and odd–even split analyses distinguish power shared by independent data subsets from split-dependent contributions, which is particularly important at high multipoles.

The workflow also differs computationally from repeated cosmological parameter inference. SkyPi does not search for a new cosmological parameter vector whenever a Planck input or sampling resolution is changed, and it does not invoke a native CPTG perturbation evolution. Its repeated tasks are primarily map ingestion, instrument and mask handling, spherical-harmonic operations, and evaluation of the registered comparison spectrum. This removes parameter-search work from each analysis while leaving high-resolution map processing computationally demanding.

Angular resolution is a scientific variable in these tests. A HEALPix sky at resolution NSIDE N contains $12N^2$ pixels. Increasing NSIDE permits finer angular structure to be represented, while a separately declared maximum multipole $\ell_{\max}$ determines which harmonic scales enter a calculation. SkyPi therefore reports both NSIDE and multipole coverage: agreement within a fixed acoustic band can be distinguished from residual changes caused by extending the analysis to smaller angular scales.

The earlier CPTG CMB comparison-map study established the initial phase-conditioned real-sky protocol and its observational and null-envelope controls [3]. The present work develops that geometric construction into a reproducible, high-resolution analysis of Planck temperature maps, with emphasis on angular-power correspondence, split spectra, temperature statistics, map residuals, and resolution dependence. The following sections first define the geometric and CMB comparison inputs, then describe the measurement operators and software before presenting the observational results.

2 From the Original CMB Map-Closure Paper to SkyPi

2.1 Original geometric field-coordinate closure

The predecessor study is not a preliminary interface demonstration; it defines the experimental problem inherited by SkyPi. Its declared primary observable was *comparison-map closure*: a locked angular-power envelope was translated into a real measured CMB harmonic field and assessed with the same map-residual metric as the corresponding Planck comparison baseline [3]. The protocol used measured harmonic phases as the spatial coordinate, then tested the permitted amplitudes. That same-phase construction does not claim to originate the positions of individual Planck hot and cold spots from the Pi constants alone. The discriminating question is whether the registered CPTG envelope maintains agreement when held fixed across real map products and explicit null envelopes.

The predecessor's primary working resolution was NSIDE 256 with $\ell_{\max} = 767$ under its documented $3N_{\mathrm{side}} - 1$ convention. For the primary SMICA result it reported a Pearson observed-field correlation of 0.983434 and a fitted residual RMS of 18.218928 μK. Across five Planck component products, the *difference between the CPTG and Planck benchmark residual RMS values* averaged 0.023518 μK; this small *difference of residual metrics* must not be confused with a 0.023518 μKobserved-minus-CPTG map residual. Generic flat, power-law and shuffled-spectrum controls fared substantially worse under the same phase scaffold; the shuffled-envelope control produced roughly 95.461 μKresidual RMS, compared with approximately 18.23–18.25 μKfor the Planck and CPTG benchmark envelopes. Table 1 records the claim level without conflating separate estimators.

Original map-closure outcome	Reported value
Primary working NSIDE / $\ell_{\max}$	256 / 767
SMICA observed-field correlation	0.983434
SMICA fitted residual RMS	$18.218928\,\mu$K
Mean CPTG–Planck residual-RMS separation, five components	$0.023518\,\mu$K
Shuffled-spectrum control residual RMS	$95.461\,\mu$K

Table 1: Foundational results from the original CPTG CMB comparison-map study [3], not a new SkyPi run. Its primary comparison used the Planck PR3 SMICA full-mission field, NSIDE 256, $\ell_{\max} = 767$, the map's own temperature mask thresholded at 0.99 (83.6431% retained), and a fitted global amplitude plus offset. Its residual estimator and mask are different from those used in the modern SkyPi rows.

2.2 The fixed geometric row replaces sector-by-sector cosmological fitting

The theory-side object is a single registered row, schematically $\Theta_\pi^{\text{locked}}$, carried to the CMB comparison coordinate by a prescribed translation $m_{\text{CMB}}(\Theta_\pi^{\text{locked}}; \nu)$, where ν represents the observation pipeline's mask, transfer, calibration and foreground conventions. The distinction from a sector-fitted theory is methodological: *observational treatment may change to match the data product; the CPTG geometric row is not reoptimized for that product.* The native Pi branch, its derived acoustic and luminosity projections, and the numerical CMB comparison interface have specific roles defined in the current Version III technical reference [1].

The numerical temperature source is evaluated from a fixed, registered comparison row rather than a row re-optimized for each Planck product. Registered low-multipole modifiers also have a documented Commander-observation conditioning history. Thus, *no per-run refit* has a specific meaning: the source is held unchanged across the analyses in this paper, while its documented historical construction remains part of the method.

2.3 What SkyPi adds to the original experiment

SkyPi extends the earlier map-level test in four ways. First, it works at native and downsampled HEALPix resolutions and records N_{side}, multipole limits and pixel/window conventions. Second, it applies explicit beam and mask operators to the fixed source before comparing it with independently measured Planck pseudo-spectra. Third, it supports common-sky HM/OE cross-power and split-difference estimators that distinguish shared signal from split-dependent power. Fourth, it produces maps, spectra, temperature statistics and recorded analysis settings that allow the comparison to be repeated. The predecessor establishes the original geometric field-coordinate claim; SkyPi demonstrates its continued operation across a much larger observational and computational domain.

3 The Fixed CPTG Geometric and CMB Comparison Coordinates

3.1 Native Pi relations

The native geometric branch is registered separately from the temperature-spectrum interface. In the notation used by SkyPi, the branch coefficients and acoustic exponent are

$$B_\pi = 1 - \frac{1}{\pi}, \qquad T_\pi = \frac{1}{\pi}, \qquad p_{\text{ac}} = 3 - \frac{\pi}{100}. \tag{2}$$

The dimensionless expansion-coordinate functions are

$$E_\pi^2(z) = B_\pi + T_\pi(1+z)^\pi, \tag{3}$$

$$E_{\text{ac}}^2(z) = B_\pi + T_\pi(1+z)^{p_{\text{ac}}}. \tag{4}$$

Here z is the redshift coordinate, E_π is the native Pi relation and E_{ac} is its registered acoustic-comparison relation. The

acoustic line-of-sight integral and Alcock–Paczynski coordinate are

$$I_{\rm ac}(z) = \int_0^z \frac{du}{E_{\rm ac}(u)}, \qquad F_{\rm AP}(z) = E_{\rm ac}(z) I_{\rm ac}(z). \tag{5}$$

The associated native curvature-radius scaling is $R_C(z) = R_{C,0}/E_\pi(z)$. These relations are implemented in `cptg_native_geometry.py`, with a registered numerical reference check of $F_{\rm AP}(2.33) \simeq 4.50808629$. They provide geometric context for the CMB workbench. They do not, by themselves, generate the numerical C_ℓ^{TT} used in the SkyPi map and spectrum comparisons.

3.2 The frozen CAMB-facing TT spectrum

The temperature-spectrum construction is held in `cptg_fixed_pi.py`. The program passes a registry of cosmological *comparison coordinates* to CAMB, which evaluates the numerical temperature angular-power spectrum [10]. This CAMB call is an external transfer calculation used in the registered CMB comparison; it is not a native CPTG perturbation or radiation-transport solver. In particular, quantities named `H0`, `ombh2` and `omch2` are arguments of the CAMB comparison interface and should not be confused with independent, adjustable constants of the native geometric equations.

CAMB-facing coordinate	**Registered value**
H_0 (comparison interface, $\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	67.4777967351
$\Omega_b h^2$	0.0225278574944
$\Omega_c h^2$ (comparison coordinate)	0.1176858416205
n_s	0.9685840734641
$N_{\rm eff}$ (`nnu`)	2.9685840734641
A_s	$2.13628300444 \times 10^{-9}$
τ	0.0589308759343
$Y_{\rm He}$, frozen PRIMAT input	0.2457602469334
$A_{\rm lens}$, Ω_k, $\sum m_\nu$ (eV)	1.0, 0.0, 0.06

Table 2: Frozen temperature-spectrum comparison coordinates in the audited SkyPi source. The names follow the CAMB numerical interface. They are held constant when observational FITS files or processing settings are changed and are not estimates obtained by fitting the individual runs presented here.

The current workbench uses a frozen helium-4 mass fraction $Y_{\rm He} = 0.24576024693340703$, recorded as a CPTG PRIMAT gate-off result in the accompanying source provenance [5]. This replaces the historical Pi Version II CMB-facing helium entry $Y_{\rm He} = 0.2485$ while leaving the other comparison coordinates fixed. The change is an explicitly documented source-input substitution, not an operation performed in response to the Planck temperature maps analyzed below. The program asks CAMB for raw C_ℓ in $\mu\mathrm{K}^2$ and applies no fit of these parameters to a selected observed sky.

3.3 Frozen low-multipole comparison factors

SkyPi's current combined source carries an additional registered set of low-multipole power factors. Writing $C_\ell^{\pi,\rm CAMB}$ for the helium-updated comparison spectrum, the source supplied to subsequent measurement operators is

$$C_\ell^{\rm SkyPi,src} = q_\ell^2 C_\ell^{\pi,\rm CAMB}, \qquad q_\ell^2 = \begin{cases} 0.1311364911405572, & \ell = 2, \\ 0.7738684053212608, & 3 \le \ell \le 9, \\ 0.8499508217809941, & 10 \le \ell \le 29, \\ 1, & \ell \ge 30. \end{cases} \tag{6}$$

These are *power* factors, not amplitude factors to be squared a second time. The source code `skypi_combined.py` applies them only to the indicated multipoles and verifies that source power at $\ell \ge 30$ is unchanged. Their provenance is a frozen Commander HM1×HM2 observational comparison dated 31 October 2026. Accordingly, "frozen" means that these values

are not recalculated for the current map; it does not mean that they were obtained without observational conditioning. SkyPi records the frozen source and its observational-conditioning provenance in the generated run metadata.

This distinction is important for interpreting cross-product correspondence. A new SEVEM map does not refit the low-ℓ factors, but the original Commander comparison is part of their documented development history. The numerical comparisons in this paper are not represented as statistically disjoint calibration and validation samples without further provenance analysis.

4 SkyPi Software Architecture

4.1 The local workbench

SkyPi combines a browser interface with Python modules running on the user's own computer. The browser presents four principal workspaces: *CMB Setup*, *CPTG Results*, *Diagnostics & Files*, and *Geometry & Phase*. A scientific worker reads the selected FITS inputs and carries out the numerical steps; the interface displays progress, results, figures and exports. The scientific computation is separated from browser presentation so that interface changes do not alter the fixed CPTG calculation.

The Setup workspace groups the observational input, unit interpretation, resolution, multipole limit, mask, beam response, and optional split processing. A checked-run procedure records the selected arguments before execution. The Results workspace displays the raw input, the processed observed map, the SkyPi comparison map, their difference and power spectra. Diagnostics & Files exposes the selected measurement operators, numerical summaries, processing messages and evidence exports. Geometry & Phase presents harmonic-amplitude diagnostics that are distinct from the optional observed-phase map visualization.

4.2 Three distinct calculation layers

The software can be read as three connected but non-interchangeable layers. The first is the *geometric registry*: the native Pi relations and frozen CMB comparison coordinates. The second is the *observation and measurement layer*: Planck map ingestion, calibration-unit handling, temperature-pixel validity, monopole and dipole conventions, masks, transfer functions and harmonic transforms. The third is the *evaluation layer*: matched spectra, split estimators, map residuals, figures and numerical diagnostics.

Layer	**Principal modules and recorded responsibility**
Geometry and frozen TT source	`cptg_native_geometry.py`; `cptg_fixed_pi.py`; `skypi_combined.py`.
Planck input and response	`raw_cmb.py`; `planck_input_response.py`; `commander_response.py`; split, mask and validity modules.
Scientific diagnostics and evidence	`phase_geometry_diagnostics.py`; temperature statistics; result CSV and JSON outputs.
Browser and execution	`skypi_browser_ui.html`; `browser_workbench.py`; `scientific_worker.py`; preflight and launcher modules.

Table 3: Functional separation of the SkyPi implementation into geometry, Planck measurement handling, diagnostics and execution. The registered geometric relations are held fixed across the reported comparisons.

4.3 Input validation, runtime boundaries and output storage

A scientific result is tied to the FITS input identity, selected setup, source-code registry and execution record. The current workbench checks recognized Planck input families, selected FITS field and unit, sampling resolution, response multipole coverage, masks and split requirements. In recognized official temperature FITS files, the temperature-unit header is authoritative; ambiguous or conflicting unit declarations are rejected rather than silently reinterpreted. Invalid temperature pixels are recorded, and admissible exclusion is treated as an explicit measurement operation rather than a modification of CPTG geometry.

SkyPi generates run results and optional full-resolution scientific FITS exports in separate local directories. The full-map FITS export is optional because a high-resolution map can be large; settings-linked filenames help identify existing results and avoid unnecessary duplicate computations. Each run produces numerical tables, a `results.json` record and scientific

figures. Readers can reproduce the reported results directly in SkyPi from the specified Planck observations and processing settings.

The browser includes a Reset to Defaults control. It changes active setup and display selections, not the registered geometric source, saved evidence or server token. Release-candidate testing separately exercises asynchronous input selection and result-status polling; those interface tests do not replace a native scientific analysis of Planck data.

5 Planck Observations and Measurement Operators

5.1 Data families and two representative analyses

The fixed CPTG geometric construction is not specific to four component-separation algorithms. SkyPi recognizes the Planck PR3 component-separated families Commander, SEVEM, SMICA, SMICA-NOSZ and NILC; it also processes supported Commander PR4, foreground-subtracted SEVEM frequency products and HFI frequency observations, with separate observational handling for 545 GHz Galactic dust [6, 7]. The completed tests considered here include full-mission, HM/OE and high-resolution configurations. The four commonly masked PR3 OE products provide a controlled *subset* for detailed cross-spectrum analysis, not the entire observational test program. Foreground-subtracted SEVEM products are component-separated frequency estimates, whereas raw HFI measurements retain substantial frequency foregrounds; their beam and residual statistics are not directly interchangeable.

Observation category	**Products examined**	**Role in this study**
PR3 CMB components	Commander; SEVEM; SMICA; SMICA-NOSZ; NILC	Full-mission and/or OE/HM CMB comparisons
PR4 component separation	Commander at NSIDE 256–4096	Same-source resolution sequence
PR3 foreground-subtracted frequency maps	SEVEM 70, 100 and 217 GHz	Frequency-specific CMB comparisons
PR4 HFI frequency maps	100 and 217 GHz full-frequency products	Raw temperature and response checks
PR3 545 GHz HFI	Full-mission surface brightness	Observed Galactic dust, not a CMB map closure test

Table 4: Breadth of Planck products examined in the SkyPi work described here. Product and instrument families are distinct from the four deliberately matched OE case studies. Frequency-product results use their own masks, beams and estimators; a 545 GHz dust-brightness analysis must not be read as a recovered CMB temperature comparison.

Recorded configuration	**SEVEM 70 GHz**	**SEVEM 217 GHz**	**Definition**
Input release	PR3 FGsub	PR3 FGsub	Component-separated temperature
Observed input	OE1+OE2 coadd	Full mission	Primary comparison sky
Analysis NSIDE	1024	2048	HEALPix resolution
$\ell_{\max}$	2048	4096	Harmonic cutoff
Retained region	75.9100%	77.9412%	Evaluated mask fraction
Nominal response choice	13.08 arcmin	4.90 arcmin	Approximate channel beam

Table 5: Run configuration for the two representative SkyPi analyses. The 70 GHz calculation uses an odd-even valid-region mask intersected with the common intensity mask and a 1-degree C2 apodized harmonic analysis window; the 217 GHz calculation uses the original common intensity mask. The nominal channel beam selections are *not* verified SEVEM effective beam transfers.

The 70 GHz primary sky is an equal-weight harmonic coadd of OE1 and OE2 maps; its full-mission file contributes header and response information but not the primary temperature pixels. The 217 GHz case uses full-mission temperature pixels directly, without selecting an HM or OE primary analysis. Both analyses have their map and processing roles recorded in `results.json`. These differences must be retained when interpreting their residual RMS values.

5.2 Map sampling, harmonic analysis and masks

A temperature field $T(\hat{n})$ on the sphere can be expanded as

$$T(\hat{n}) = \sum_{\ell=2}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n}), \qquad C_\ell = \frac{1}{2\ell+1} \sum_m |a_{\ell m}|^2. \tag{7}$$

The sum starts at $\ell = 2$ for the band-limited CMB comparison after the run's monopole/dipole convention. SkyPi uses HEALPix/healpy for spherical sampling and harmonic calculations [11]. A run's chosen NSIDE fixes the analysis grid, while the chosen $\ell_{\max}$ fixes the retained harmonic extent. The implementation provides an automatic multipole mode and explicit bounds, with the larger NSIDE choices demanding substantially more memory and computation.

For a mask or weighted analysis window $w(\hat{n})$, the measured field is wT and its direct angular-power estimate is a *pseudo-spectrum* $\widetilde{C}_\ell$, not the deconvolved full-sky C_ℓ. A matching comparison source can be forwarded through the same window with a mask-coupling matrix:

$$\widetilde{C}_\ell^{\mathrm{SkyPi,fwd}} = \sum_{\ell'} M_{\ell\ell'}[w] \left(P_{\ell'}^2 B_{\ell'}^2 C_{\ell'}^{\mathrm{SkyPi,src}} \right), \tag{8}$$

where P_ℓ denotes any included pixel-window factor, B_ℓ the selected effective amplitude-response convention, and $M_{\ell\ell'}$ the coupling appropriate to the recorded mask. Equation (8) is the operator sequence in schematic form; the actual run's pixel-window, filter and response definitions are retained in the evidence and govern how each factor is evaluated. This avoids equating a mask-coupled prediction with an unmasked theoretical curve. The pseudo-C_ℓ and coupling approach follows the established MASTER family of estimators [12].

The 70 GHz analysis records the intersection of the Planck common intensity and odd-even missing-pixel masks, followed by a C2 one-degree analysis apodization. The 217 GHz analysis records the common intensity mask directly. Final display masking is separately identified because changing which pixels are rendered can change a displayed histogram without altering the earlier harmonic source spectrum.

5.3 NSIDE and angular resolution are scientific variables

HEALPix sampling at $N_{\mathrm{side}} = N$ provides

$$N_{\mathrm{pix}} = 12N^2, \qquad \Omega_{\mathrm{pix}} = \frac{4\pi}{12N^2}. \tag{9}$$

The nominal number of pixels therefore rises from 786,432 at NSIDE 256 to 201,326,592 at NSIDE 4096, a factor of 256. SkyPi's common automatic scientific setting is approximately $\ell_{\max} = 2N_{\mathrm{side}}$ where supported, but the original comparison-map paper used $3N_{\mathrm{side}} - 1$, and any explicit user choice must be reported rather than silently equated to the automatic policy. These differences matter even when the same theoretical registry and input file are held constant.

Three distinct resolution questions must not be conflated. *Pixel sampling* determines the sky grid. *Harmonic support* determines the upper ℓ included in a map and spectrum. *Instrument response and analysis mask* determine what measured power is meaningful at each ℓ. Higher NSIDE commonly increases the amount of recoverable spatial detail; whether a measured discrepancy decreases depends on the observable, comparison interval and processing operator. In a resolution study, a common-band statistic compares the same multipoles across grids, while a full-range residual RMS deliberately incorporates newly admitted high-ℓ modes. The latter may rise even when the common-band correspondence improves.

At NSIDE 4096 the HEALPix grid resolves finer angular structure and supports an expanded multipole interval. Section 7.7 quantifies its impact on fixed-band power correspondence and on the residual statistics evaluated over each run's stated harmonic range.

5.4 Beam transfer and the NPIPE distinction

A finite instrumental beam smooths the sky, modifying measured harmonic amplitudes. When a file tabulates the amplitude response B_ℓ, the corresponding temperature auto-power response is B_ℓ^2. An already power-defined window

function must instead be applied according to its own definition; source conventions must not be squared blindly. SkyPi therefore records the response source, type and range used in a run.

The two reported case studies selected approximate 13.08-arcminute and 4.90-arcminute Gaussian channel responses. These numbers provide transparent working assumptions, but their labels do not certify the cleaned-map effective transfers. More detailed beam calculations are available in the Planck literature, including the QuickPol formalism [13]. For NPIPE/PR4, published processing and CamSpec analyses employ frequency-specific transfer functions, with noise and detector-set cross-correlations assessed separately [8, 9]. Their NPIPE A/B detector-set splits are constructed differently from the PR3 HM and OE splits; the responses and split covariance conventions should not be interchanged without verification.

An independently examined NPIPE 217×217 GHz beam FITS illustrates how a tabulated response can differ from a nominal Gaussian, particularly at high multipoles. That auxiliary response inspection does *not* establish a new SEVEM 217 GHz temperature-map residual, because the effective SEVEM transfer and covariance remain separate requirements. The results below use the instrument-response selections made in the respective SkyPi analyses.

6 The SkyPi Comparison and Diagnostic Measurements

6.1 Observed pseudo-spectrum and fixed forward power

SkyPi calculates an observed pseudo-spectrum from the processed Planck temperature field without reading the CPTG source power into that estimator. The same declared analysis window and response conventions are then applied to the fixed SkyPi source. To display power in the conventional units of $\mu\mathrm{K}^2$, the figures and tables commonly use

$$D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell, \qquad \widetilde{D}_\ell = \frac{\ell(\ell+1)}{2\pi} \widetilde{C}_\ell. \tag{10}$$

The measured masked-sky estimate and the masked-forward source occupy the same numerical comparison space, but they have distinct origins. One is derived from actual temperature pixels; the other starts with the frozen Pi comparison registry and passes through recorded operators. Band averages are formed only from the multipole interval and estimator explicitly identified in each table.

6.2 Observed-phase SkyPi map reconstruction

In addition to spectrum comparison, the software constructs a SkyPi temperature map that expresses the fixed comparison amplitudes using measured harmonic spatial coordinates. If $a_{\ell m}^{\mathrm{obs}}$ are the complex coefficients of the processed observed map, a schematic form of this operation is

$$a_{\ell m}^{\mathrm{SkyPi,map}} = a_{\ell m}^{\mathrm{obs}} \left(\frac{C_\ell^{\mathrm{SkyPi,comparison}}}{C_\ell^{\mathrm{obs}}} \right)^{1/2}, \quad \ell \geq 2, \tag{11}$$

with zero-power protection and any common harmonic filter treated as recorded in the execution. The relevant spectrum ratio changes the collective amplitude at a multipole but preserves the observed relative pattern among its m modes. Thus the resulting full-sky SkyPi temperature image retains Planck-derived angular phase information. Its hot and cold spot locations must *not* be presented as independently generated from the Pi relations alone. Instead, the image answers a useful and concrete question: how does the observed spatial structure look when expressed with the registered SkyPi harmonic amplitude distribution?

For this reason, map correlation and spectral agreement are separate descriptors. A large pixel correlation in Equation (11) is informative about the amplitude reweighting and preserved morphology, but is not an independent phase-location test. Geometry & Phase diagnostics additionally report phase-angle-independent harmonic-amplitude distance, permitting a second reading of agreement that does not assign independent origin to the retained phases.

The phase-angle-independent measurement is also visible in the 70 GHz diagnostic figure. It plots the square roots of the observed and masked-forward pseudo-power, then the squared amplitude discrepancy per multipole, all without using the observed phase angles as the object being tested. The low-ℓ peaks and gradually rising high-ℓ contribution describe where the spectral-amplitude differences occur; their plotted magnitude is not a covariance-calibrated significance.

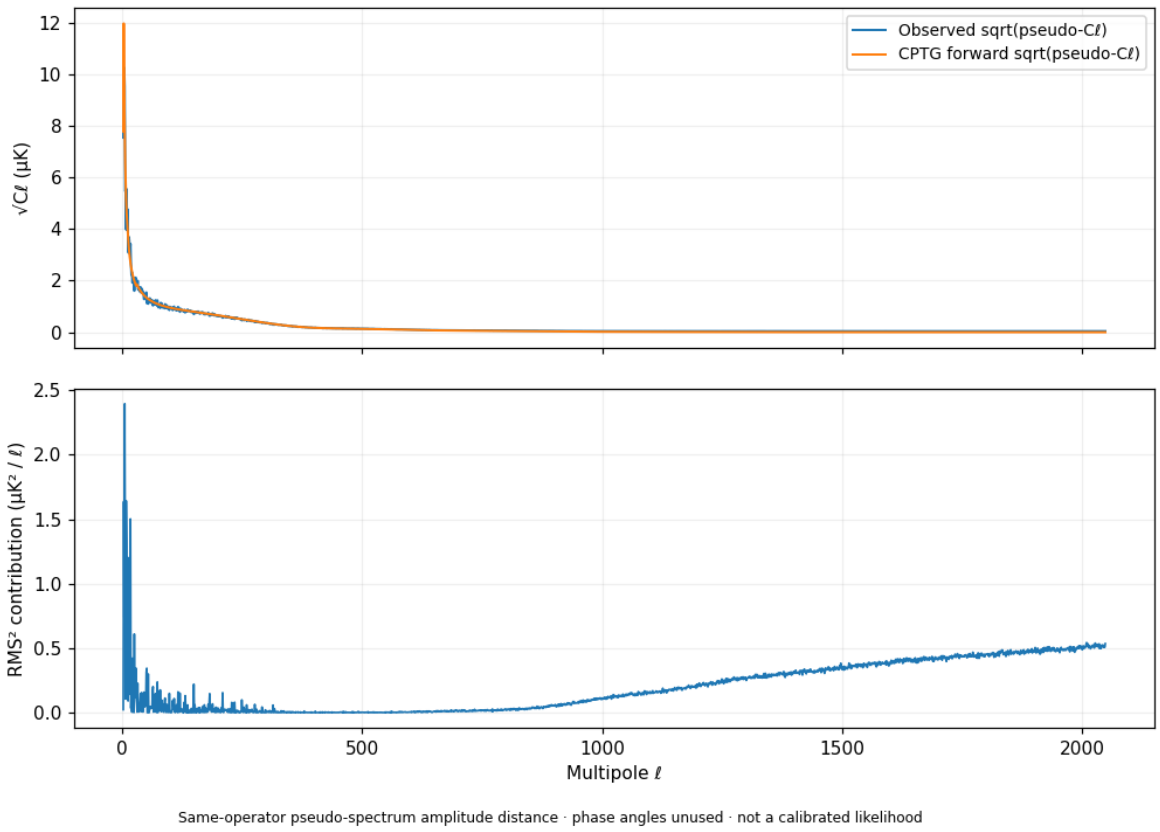


Figure 1: SkyPi's 70 GHz phase-angle-independent amplitude comparison. The upper panel compares $\sqrt{\widetilde{C}_\ell}$ in μK under the same operator; the lower panel shows the multipole contribution to squared amplitude distance. The figure does not claim that SkyPi calculates Planck's individual angular phases independently.

6.3 Pixel residuals and band differences

Let $\mathcal{R}$ be the set of retained pixels and $N_\mathcal{R}$ its size. The signed residual and its pixel RMS are

$$\Delta T_p = T_p^{\mathrm{obs}} - T_p^{\mathrm{SkyPi}}, \tag{12}$$

$$\sigma_{\Delta T} = \left(\frac{1}{N_\mathcal{R}} \sum_{p \in \mathcal{R}} (\Delta T_p)^2 \right)^{1/2}. \tag{13}$$

SkyPi also records the Pearson map correlation on the same retained region. Neither statistic is a likelihood χ^2; both include the effects of the selected map, residual foregrounds, noise, response and preprocessing. In spectrum tables, a common descriptive band contrast is

$$\delta_b = 100 \frac{\langle D_\ell^{\mathrm{SkyPi}} \rangle_b - \langle D_\ell^{\mathrm{observed}} \rangle_b}{\langle D_\ell^{\mathrm{observed}} \rangle_b} \,\%, \tag{14}$$

provided that the denominator is positive and that the two band means use matched definitions. For near-zero or negative cross-power, the ratio may become unstable or undefined and is not used as a precision indicator.

6.4 Retained-region temperature statistics and distribution closure

SkyPi exports a *Temperature Statistics* figure for every completed map analysis. The figure overlays the retained-region probability densities for the observed band-limited temperature field T^{obs}, the fixed-source SkyPi comparison field T^{SkyPi},

and their *paired-pixel* difference $\Delta T = T^{\mathrm{obs}} - T^{\mathrm{SkyPi}}$, all in μK. The distributions are plotted on a logarithmic density axis to expose both the central concentration and the hot/cold tails. They are computed over the declared retained mask region, not over the original, unprocessed FITS field; the plotted SkyPi map uses the observed harmonic phase coordinates described in Equation (11).

The near-coincidence of the two temperature distributions measures *population-level correspondence*: hot and cold temperature values appear with closely related frequencies under the registered amplitude envelope. The much narrower *observed-minus-SkyPi* distribution measures their actual, position-matched differences rather than an arbitrary difference of unrelated histograms. These are complementary tests: similar marginal histograms by themselves would not guarantee small paired-pixel residuals or angular-power agreement. We therefore report the distribution plots alongside residual RMS, pixel correlation, and the separately constructed harmonic-power measurements. A distribution figure is a descriptive physical-temperature diagnostic, not a covariance-calibrated likelihood test.

6.5 Binary sky fraction and harmonic-window power

The mask specification is part of the numerical definition of a comparison. Let M_p indicate eligibility of a HEALPix pixel and let $W_p \in [0, 1]$ be the weight applied in the harmonic operator. SkyPi's listed *retained sky fraction* is the binary pixel fraction, whereas the quadratic harmonic weight is

$$f_{\mathrm{sky}}^{\mathrm{binary}} = \frac{1}{N_{\mathrm{pix}}} \sum_p M_p, \qquad f_{W^2} = \frac{1}{N_{\mathrm{pix}}} \sum_p W_p^2. \tag{15}$$

In an OE run, the recorded M_p is the intersection of Planck's common intensity mask and the official OE missing-pixel selection, with both thresholded at 0.999. A one-degree spherical C2 quintic taper is then used for the split harmonic analysis. For the four matched NSIDE-2048 runs, $f_{\mathrm{sky}}^{\mathrm{binary}} = 0.763376$ while $f_{W^2} \simeq 0.531979$. These are not competing estimates of one number: one describes the retained map pixels and the other describes a weighted harmonic window. Full-mission runs without split apodization instead use their recorded common-mask selection; NSIDE changes can alter the eligible pixel set through conservative downgrade or parent replication.

This separation matters because a change in either the mask boundary, the fraction of valid sky, or the analysis-window weights changes the masked pseudo-spectrum and the retained-region residual, even with identical CPTG source amplitudes. Numerical agreement reported under one masking procedure is not to be assigned to another without recalculation. Section 8.4 specifies the operators used by each reported run.

6.6 Odd-even and half-mission splits

Where compatible observation splits are available, SkyPi can form equal-weight coadds, split cross-spectra and half-differences. For two maps A and B, define $S = (A + B)/2$ and $D = (A - B)/2$. With identical linear harmonic and mask operators applied to the pair, the quadratic identity

$$\widetilde{C}_\ell^{SS} = \widetilde{C}_\ell^{AB} + \widetilde{C}_\ell^{DD} \tag{16}$$

expresses the coadd power as shared cross-power plus difference power (with the real part used for harmonic cross-products). It is a bookkeeping identity under the matched conventions, not a claim that D is composed entirely of instrumental noise. Shared foregrounds, split-specific residuals and correlations can all affect its physical interpretation.

The split cross-spectrum is particularly valuable at high multipoles when a coadd auto-spectrum is large but the shared cross signal is small. Published Planck analyses also use split differences and simulations to estimate and verify noise, and document important differences between PR3 and NPIPE noise construction [9]. SkyPi records the split identity and its diagnostic components while withholding covariance-certified significance when the appropriate effective transfer and noise model have not been supplied.

7 Observational Results: Geometric Correspondence Across Products and Resolution

7.1 Measured values and reconstruction configurations

Every map correlation, residual RMS, temperature distribution and power-band percentage in this section is conditional on the *selected observational field, resolution, multipole range, sky mask, mask resampling, edge window, beam response and split estimator*. These choices enter the measurement operator and can materially alter the result without changing the CPTG geometric source. Accordingly, run identifiers printed with the numerical tables and figure descriptions refer to the complete settings registry in Table 15. Reported sky fractions are calculated from the final binary retained-pixel set, before any harmonic apodization; they do not by themselves specify the weighted harmonic window.

7.2 Historical baseline and new test hierarchy

The original NSIDE-256 map-closure analysis established the field-coordinate result and discrimination against its tested generic power envelopes (Table 1). SkyPi's newer results are intentionally *not* presented as the same estimator repeated at higher resolution: newer maps use no per-run amplitude/offset fit, and separate masked-forward harmonic and HM/OE split estimators. Their role is to extend the fixed-geometric solution into additional observational coordinates with explicit measurement conventions.

7.3 Four Planck products at NSIDE 2048: one fixed geometric row

The Planck PR3 OE analyses of Commander, SEVEM, SMICA and NILC (P1–P4) use NSIDE 2048, $\ell_{\max} = 4096$, and the same official common-intensity/OE-validity binary intersection, retaining 76.3376% of the analyzed sky before the one-degree C2 split-analysis taper. The four analyses have different family-specific beam operators. The reported RMS and correlations are conditional on this complete setting. Each produces a separate observed/SkyPi/residual map using the frozen Pi source, without local parameter optimization. All four analyses hold the CPTG geometric source and the declared mask treatment fixed, while retaining each Planck product's distinct observational identity. This is a much more comparable cross-product test than juxtaposing a masked low-frequency split run with an unrelated full-mission high-frequency run.

PR3 OE product / run	Binary sky	Residual RMS (μK)	Map correlation
Commander (P1)	76.3376%	9.9281	0.993801
SEVEM (P2)	76.3376%	10.2111	0.993454
SMICA (P3)	76.3376%	9.7941	0.993964
NILC (P4)	76.3376%	8.8105	0.995103

Table 6: Four PR3 OE analyses at NSIDE 2048, $\ell_{\max} = 4096$, with the same Planck common-intensity and OE missing-pixel mask intersection at threshold 0.999 and a 1° C2 apodized split-analysis window. The displayed percentage is *binary* retained sky; the weighted window is described in Section 6.5. The response differs by family (P1–P4 in Table 15). No per-run fixed-Pi parameters were adjusted.

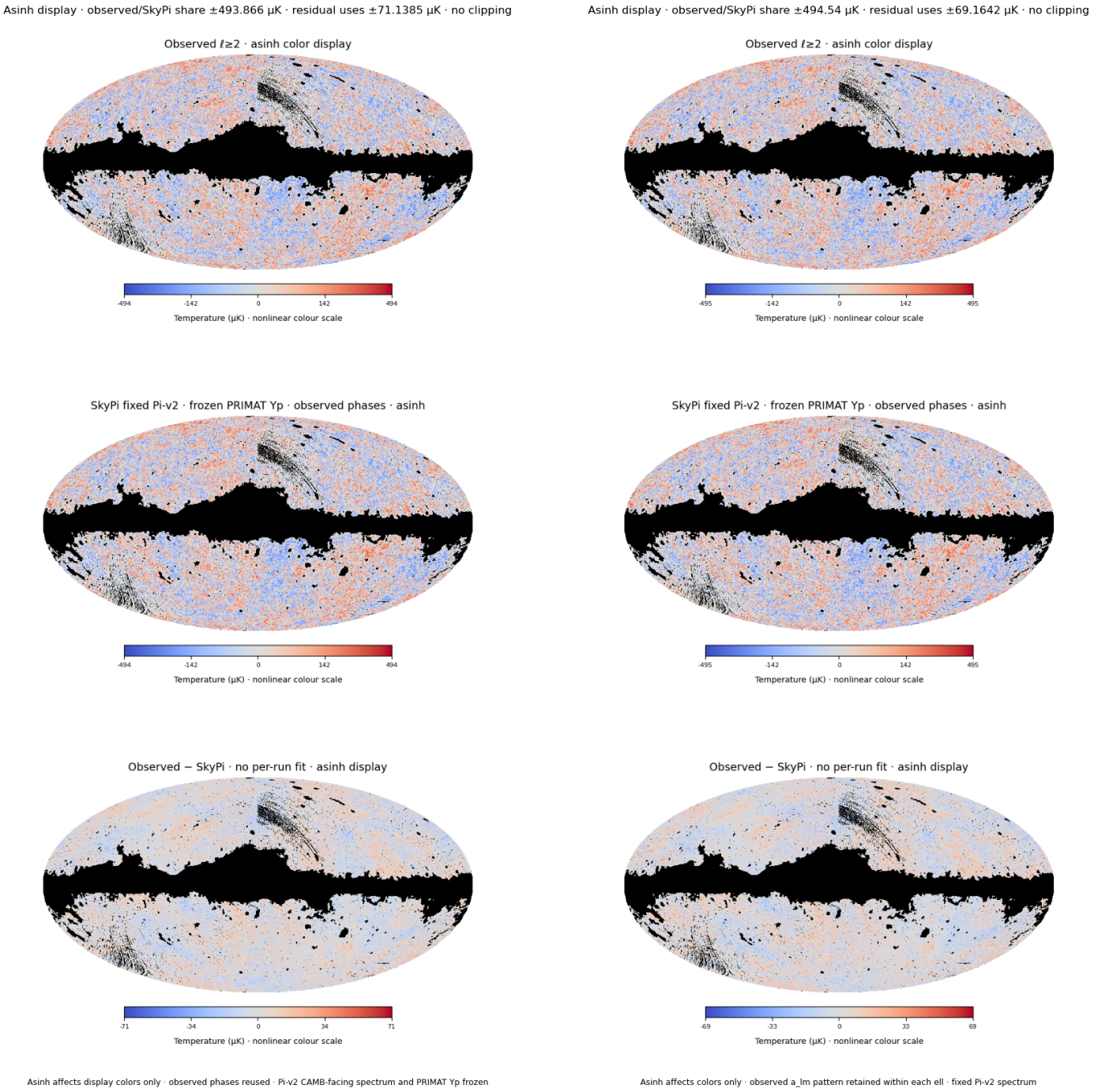


Figure 2: Runs P1 and P2: Planck PR3 OE map triptychs at NSIDE 2048, using common∩OE binary masking at 0.999 and one-degree C2 split apodization: Commander (left) and SEVEM (right). Within each triptych: observed sky, SkyPi map with registered geometry and measured phase scaffold, and observed minus SkyPi residual. The mask and high-contrast display conventions are common to the two runs.

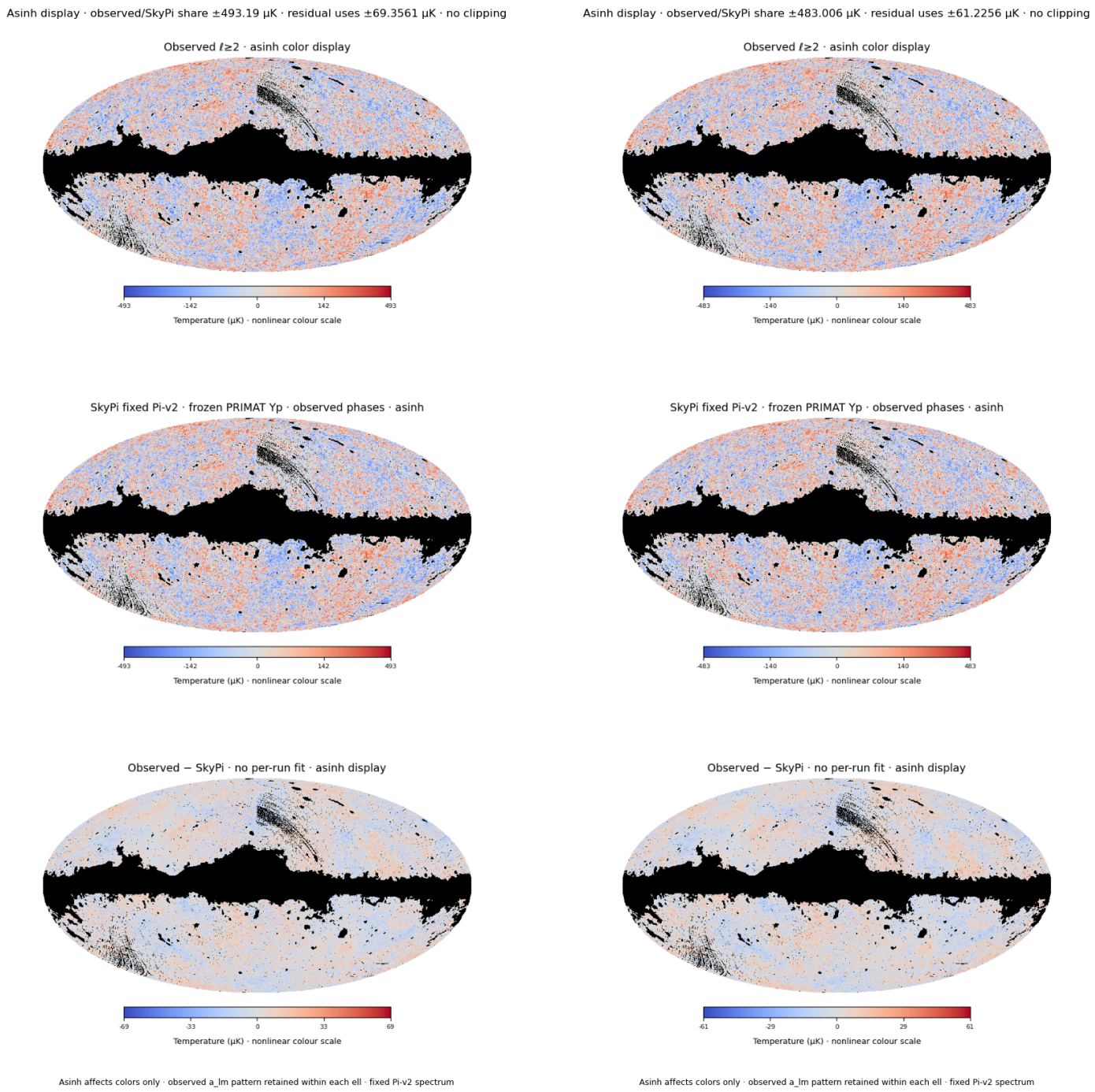


Figure 3: Runs P3 and P4: the NSIDE-2048 processing arrangement for SMICA (left) and NILC (right), with the common∩OE mask and one-degree C2 split window. Cross-product morphological stability is shown visually; the primary quantitative spectral comparison is reported separately in Table 8.

7.4 Temperature-statistics consistency across four Planck pipelines

The four NSIDE-2048 PR3 OE products (P1–P4) permit a controlled comparison of the same diagnostic under their common binary mask, one-degree apodized split window and fixed-source settings. Figures 4–5 show the temperature-statistics graphics generated by SkyPi. In each, the Planck and SkyPi temperature distributions track each other closely across positive and negative tails, while the paired-pixel residual is concentrated about zero. The figures are included *in addition to*, not instead of, the masked-forward OE cross-spectrum comparisons in Table 8.

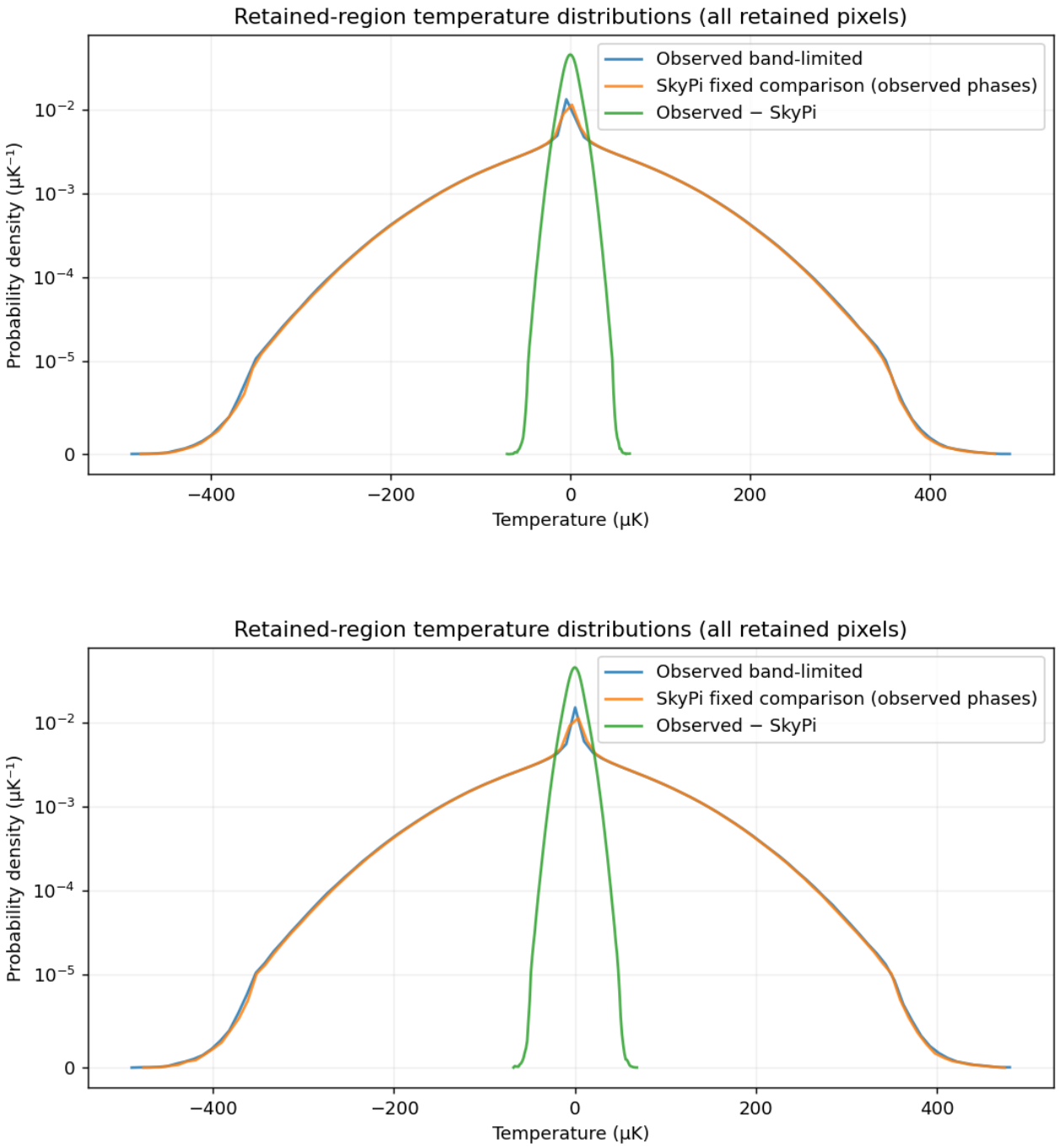


Figure 4: SkyPi temperature-statistics distributions at NSIDE 2048 and $\ell_{\max} = 4096$: Commander OE (upper) and SEVEM OE (lower). Both runs retain 76.338% sky with the common recorded mask construction. Blue and orange show observed and fixed-source SkyPi temperature populations; green shows the paired observed-minus-SkyPi residual. Original SkyPi plots are reproduced without modifying their data or palettes.

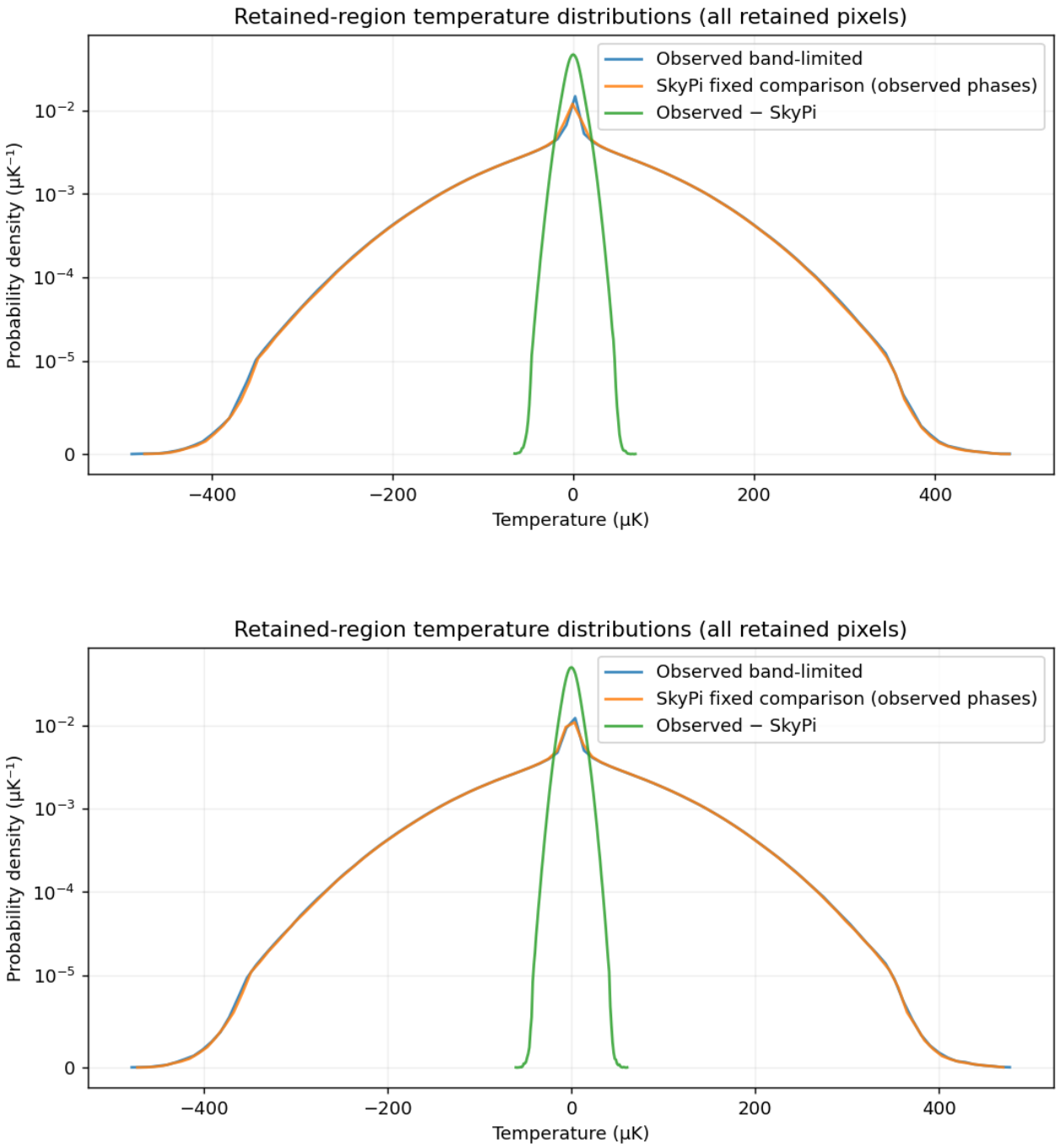


Figure 5: The corresponding temperature-statistics distributions for SMICA OE (upper) and NILC OE (lower), using the same analysis resolution, upper multipole and retained mask. These are physical temperature distributions rather than map-display contrast histograms; matching visual shape and a narrow paired residual jointly characterize the map-domain correspondence.

OE product	**Observed RMS (μK)**	**SkyPi RMS (μK)**	**Residual 5–95% (μK)**
Commander	89.2967	88.8662	$[-16.54, +16.43]$
SEVEM	89.3810	88.8706	$[-17.11, +16.83]$
SMICA	89.2603	88.8751	$[-16.36, +16.15]$
NILC	89.1026	88.8818	$[-14.67, +14.40]$

Table 7: Physical-temperature statistics generated by SkyPi for the four NSIDE-2048 OE runs. The residual intervals are actual 5th and 95th percentiles, not uncertainties on the spectrum or confidence intervals for the underlying cosmology. The same runs' residual RMS and map correlation are reported in Table 6. The reported values are directly reproducible with the stated SkyPi configuration.

The correspondence is not limited to a single component-separated map: all four fixed-source reconstructions produce signal RMS values between 88.866 and 88.882 μK, compared with observed RMS values between 89.103 and 89.381 μK. This repeated temperature-population agreement, together with the narrow position-matched residuals and the independent split cross-power bands, strengthens the multi-observable observational correspondence argument. The distribution plots alone cannot identify the causal origin of the fixed-spectrum agreement or replace a noise-covariance study.

The independently processed OE cross-spectra offer a complementary, phase-angle-independent view of the same frozen source. Table 8 reports selected percent differences between SkyPi masked-forward D_ℓ and the corresponding Planck OE1×OE2 cross band means. Around the acoustic multipoles 450–799, the four products occupy a narrow pattern of positive deviations approximately 1–4%, with the exact value depending on the multipole interval. This does not use pixel-map correlation as the evidence for spectral agreement.

Band ℓ	**Commander**	**SEVEM**	**SMICA**	**NILC**
450–549	+1.10%	+1.15%	+1.14%	+1.01%
550–649	+1.52%	+1.57%	+1.70%	+1.46%
650–749	+2.60%	+2.67%	+3.02%	+2.68%
750–799	+3.54%	+3.46%	+3.86%	+3.46%
800–999	+2.80%	+2.80%	+3.29%	+2.87%
1000–1499	+3.73%	+3.94%	+4.48%	+4.31%

Table 8: SkyPi minus observed PR3 OE cross-spectrum band-mean power for runs P1–P4 (Table 15), expressed relative to positive OE cross-power. All use NSIDE 2048, $\ell_{\rm max} = 4096$, common∩OE masks at threshold 0.999, 76.3376% binary sky and a 1° C2 split-analysis window. Family-specific full-mission beam proxies differ and split transfers/covariances are not independently certified.

7.5 Additional Planck-family temperature tests

The matched four-pipeline comparison is supplemented by completed SkyPi analyses of SMICA-NOSZ, foreground-subtracted 100 and 217 GHz OE maps, and full-mission Commander maps at other resolutions; their configuration identifiers P5, F1, F2, C1, C2 and N4 specify the measurement operators in Table 15. Table 9 reports the retained-region map statistics directly for their respective processing configurations. Unlike Table 6, these entries do not all share one mask, one multipole range or one split configuration; they establish broader observational coverage and are not ordered as a ranking of geometric accuracy. Each run retains the registered source geometry without fitting a new CMB envelope to its selected observation.

Product / run	**NSIDE**	**$\ell_{\rm max}$**	**Binary sky**	**RMS (μK)**	**Corr.**
SMICA-NOSZ PR3 OE (P5)	2048	4096	76.3376%	10.2849	0.993392
SEVEM FGsub 100 GHz OE (F1)	2048	4096	76.3376%	17.0306	0.980597
SEVEM FGsub 217 GHz OE (F2)	2048	4096	76.3376%	12.1858	0.990810
Commander PR3 full (C1)	1024	3071	77.7964%	8.4089	0.996908
Commander PR3 full (C2)	2048	4096	77.9412%	10.8978	0.994834
Commander PR4 full (N4)	4096	8192	77.9412%	17.3976	0.987254

Table 9: Additional retained-region, unfitted map comparisons. Runs P5/F1/F2 use the binary common∩OE mask and one-degree C2 analysis apodization; C1/C2/N4 use the common-intensity mask alone without OE apodization. The exact mask resampling and beam choices are given by run ID in Table 15. Differing masks, beams, resolutions, and harmonic intervals preclude ranking these residuals as equivalent experiments.

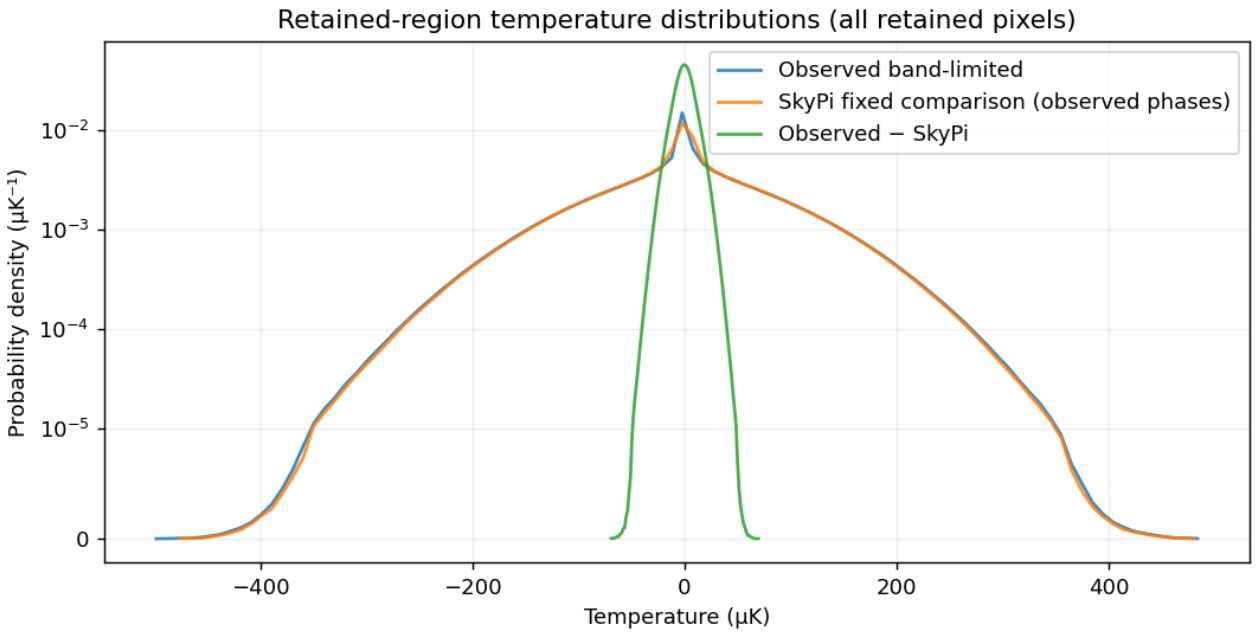


Figure 6: SkyPi Temperature Statistics for PR3 SMICA-NOSZ OE run P5 (NSIDE 2048, $\ell_{\max} = 4096$; common∩OE mask, 76.3376% binary sky, one-degree C2 window and 5-arcmin nominal response). The observed and SkyPi temperature populations are shown together with the narrower paired residual distribution.

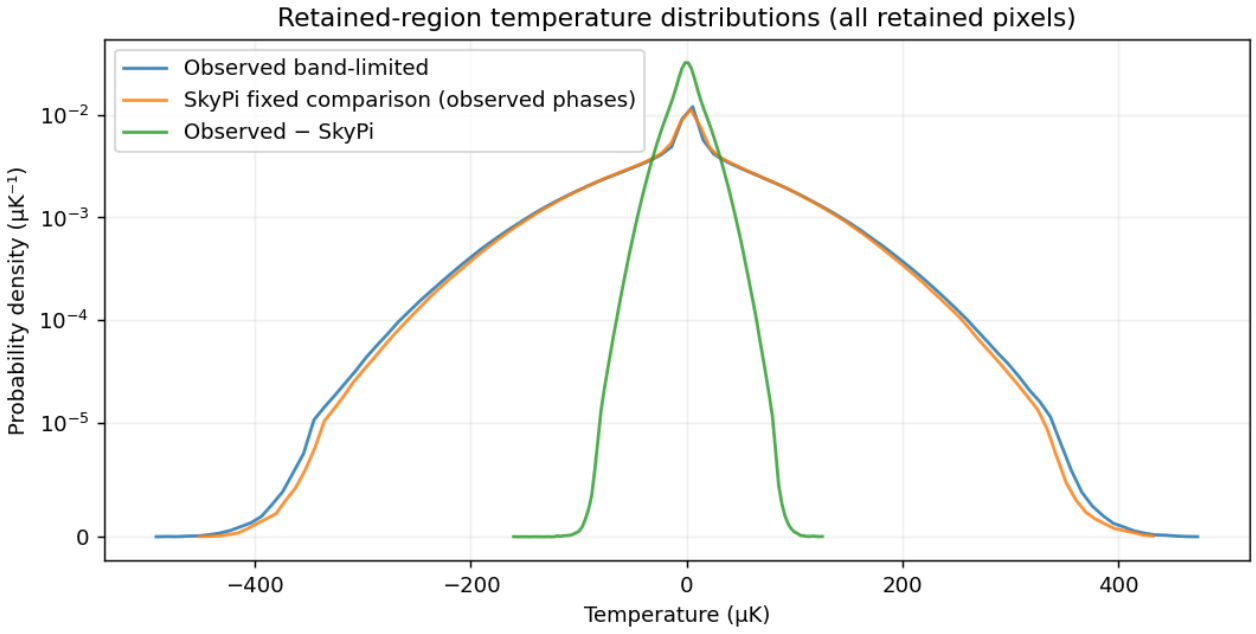


Figure 7: SkyPi Temperature Statistics for PR3 SEVEM foreground-subtracted 100 GHz OE run F1 at NSIDE 2048 and $\ell_{\max} = 4096$ (76.3376% binary sky, common∩OE mask, one-degree C2 window, 9.66-arcmin nominal response). Its higher residual relative to the four matched component products is a measured feature of this frequency-specific processing case, not a revised geometric fit.

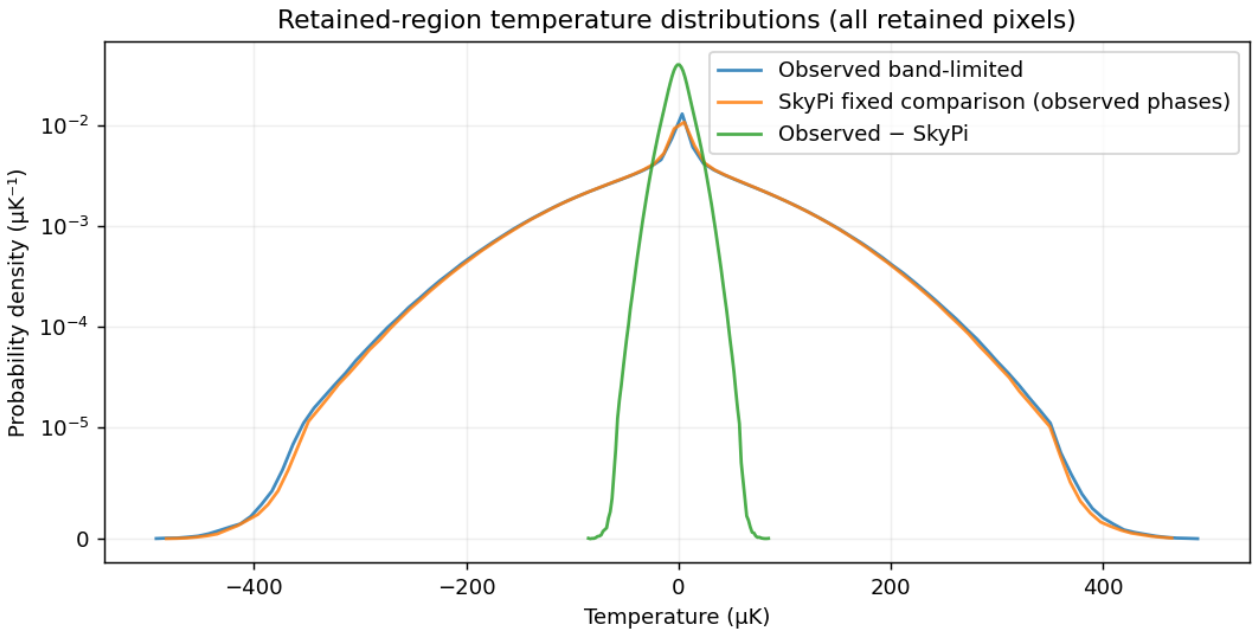


Figure 8: Temperature Statistics for the PR3 SEVEM foreground-subtracted 217 GHz OE run F2 (common∩OE mask, 76.3376% binary sky, one-degree C2 taper), distinct from full-mission run F4 with 77.9412% binary sky and no split apodization in Figure 14. These two settings select different primary observed fields and masks and give different residual statistics.

Under its explicitly different mask and split estimator, the additional 217 GHz OE run gives 12.1858 μK residual RMS and correlation 0.990810, compared with 13.4411 μK and 0.99231 for the separately configured full-mission 217 GHz case. This is evidence of distinct observational estimators, not a before–after modification of the fixed CPTG geometry. The SMICA-NOSZ entry adds another component-separation variant, and the 100 GHz OE result extends frequency-specific testing beyond the 70 and 217 GHz case studies. Temperature-distribution agreement and acoustic power must still be evaluated as distinct observables.

7.6 Raw HFI and Galactic dust: measurement scope beyond cleaned CMB

SkyPi has also processed PR4 HFI full-frequency maps at 100 and 217 GHz. The recorded results in Table 10 document frequency-map processing and response checks. These foreground-containing measurements are not foreground-cleaned component-separated CMB products, and their residual and correlation statistics are not counted as CMB closure evidence.

PR4 HFI product / run	NSIDE/$\ell_{\max}$	Binary sky	RMS (μK)	Corr.
100 GHz full (H1)	2048 / 4096	77.9412%	22.5646	0.977797
217 GHz full (H2)	2048 / 4096	77.9412%	70.9811	0.929915

Table 10: Full-frequency measurement-workflow tests (H1 and H2 in Table 15), not cleaned-CMB closure tests. Both use common-intensity masking at 0.999, with no OE splits or C2 apodization. Their nominal response assumptions differ from those of the component-separated products.

A separate observed-only 545 GHz HFI analysis at NSIDE 2048 and $\ell_{\max} = 4096$ measures Galactic surface brightness in MJy sr^{-1}, not reconstructed CMB temperature in μK. The observed surface-brightness RMS is 13.3155 MJy sr^{-1}. There is no CPTG CMB map correlation or temperature residual for that dust-analysis mode. Separating these observation classes is essential to the proper interpretation of SkyPi's broad Planck coverage.

7.7 The NSIDE sequence: what improves, what stabilizes, and what becomes visible

The Commander PR4/NPIPE runs N1–N4 use the *same* source temperature FITS, the same native Planck common-intensity mask file, the same published nominal 5-arcmin response and the same fixed CPTG geometric source. The

analysis NSIDE and associated $\ell_{\max}$ change, and conservative mask resampling changes the retained-pixel fraction. This is a fixed-source resolution series, *not* a matched-mask, fixed-multipole test; the mask fractions and harmonic limits must be read together with its band results.

Run / NSIDE	$\boldsymbol{\ell_{\max}}$	**Binary sky**	**RMS (μK)**	**Corr.**	$\boldsymbol{\lvert\delta_{450-511}\rvert}$
N1 / 256	512	76.6440%	6.4610	0.997896	2.565%
N2 / 512	1024	77.4641%	6.4925	0.998109	1.096%
N3 / 1024	2048	77.7964%	7.3824	0.997638	1.096%
N4 / 4096	8192	77.9412%	17.3976	0.987254	1.104%

Table 11: Same Commander PR4 source and nominal 5-arcmin response for all four runs (N1–N4). The official NSIDE-2048 common intensity mask is thresholded at 0.999, then conservatively downgraded for N1–N3 or parent-replicated for N4. The *different actual retained sky fractions* are listed explicitly. Common-band $|\delta|$ covers $\ell = 450$–511; the reported full-map residuals use each row's distinct $\ell_{\max}$ and mask. Neither variation is a change in the fixed geometry.

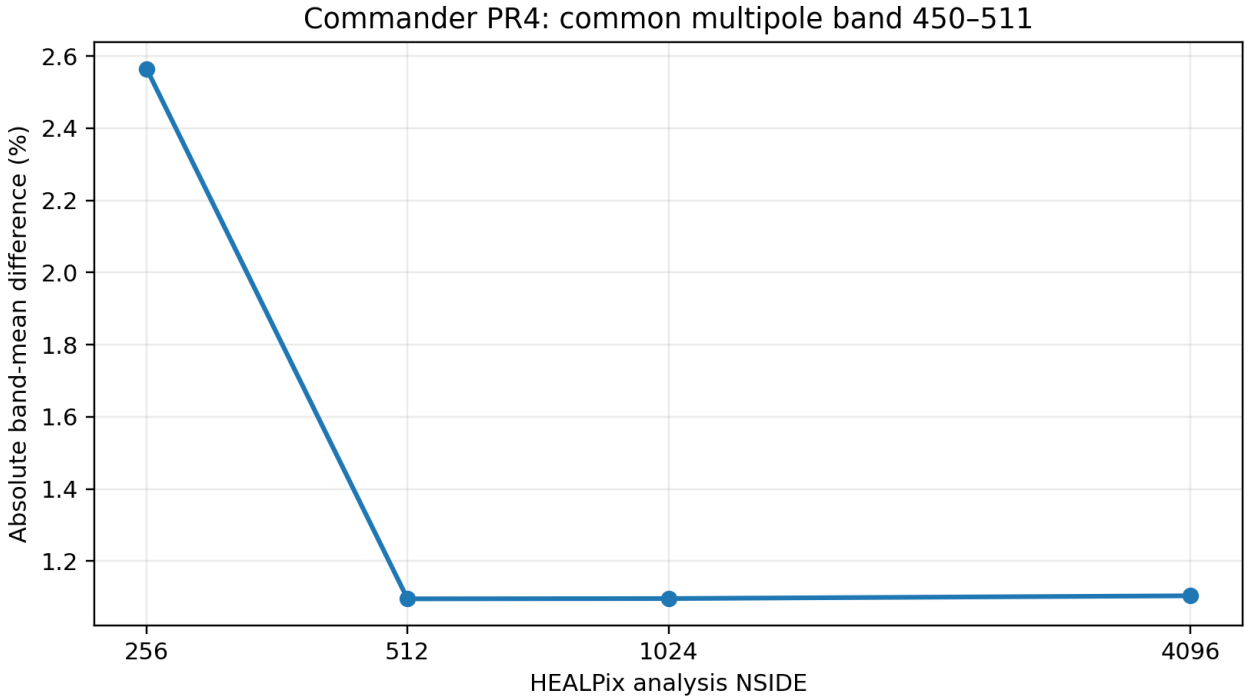


Figure 9: Absolute difference between the fixed SkyPi and observed masked band-mean D_ℓ in the same multipole interval $450 \leq \ell \leq 511$. The large improvement from NSIDE 256 to 512 is followed by a near-plateau in this fixed band; the result comes from the same Commander PR4 input and registered fixed geometry.

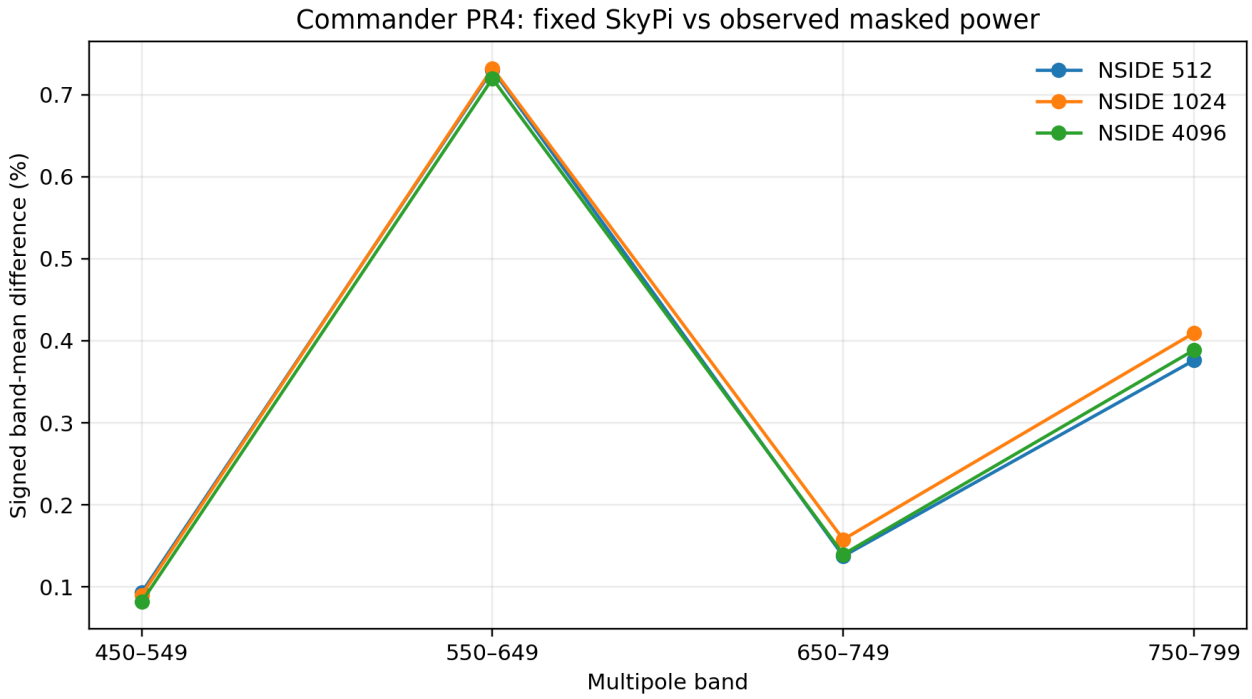


Figure 10: Signed acoustic band-mean differences for the three Commander PR4 resolutions that contain the full 450–799 range. The fixed geometry remains unchanged. A small, resolution-dependent movement is present, but the higher grids give nearly superposed values in these bands.

The detailed bands are informative. At NSIDE 512, 1024, and 4096, the respective percent discrepancies in 450–549 are approximately +0.093, +0.090, and +0.082%; in 550–649 they are +0.730, +0.732, and +0.720%; in 650–749 they are +0.137, +0.158, and +0.140%; and in 750–799 they are +0.376, +0.410, and +0.389%. The resulting band differences are stable to a few hundredths of a percentage point across these three analysis resolutions. Increased NSIDE additionally permits progressively finer angular structure and higher multipoles to enter the reconstruction.

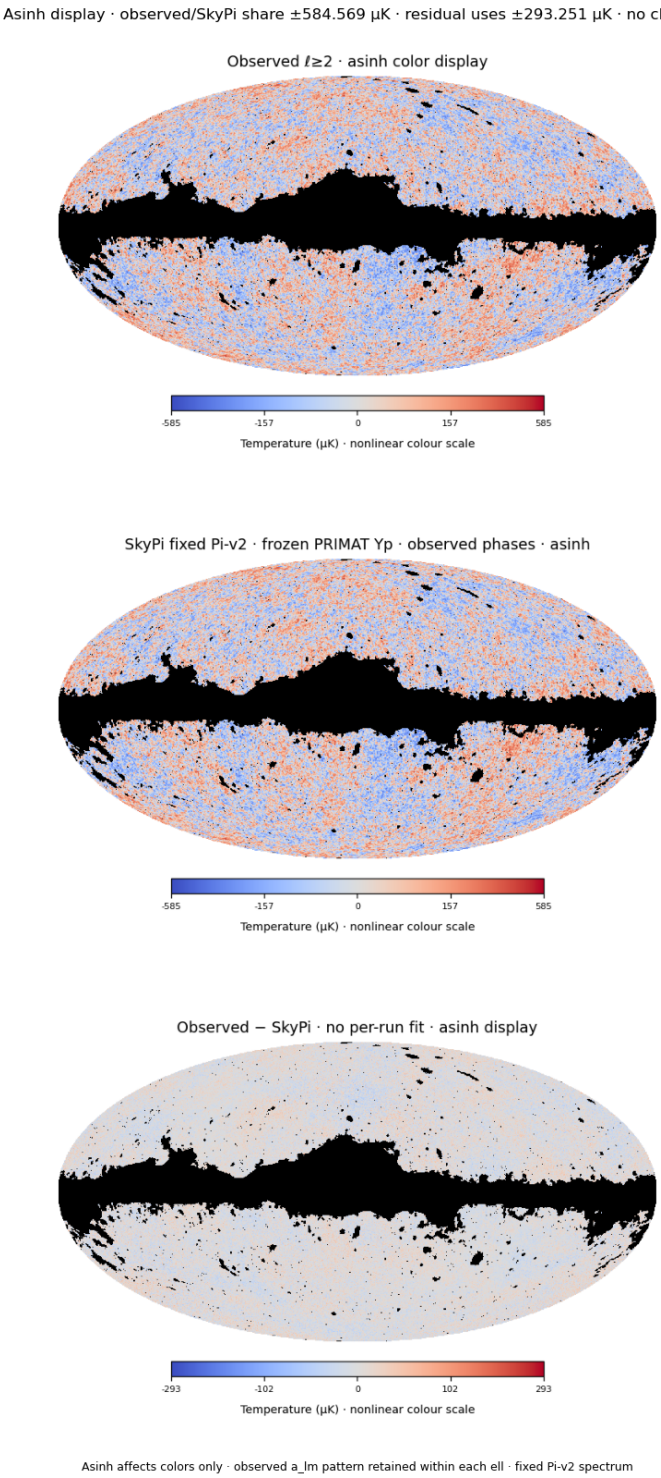


Figure 11: Commander PR4 NSIDE-4096, $\ell_{\max} = 8192$ comparison. The image and Table 11 distinguish full-range from matched-band observables: high-resolution structure is added to both observed and residual maps, and the full-range residual includes those newly admitted modes. The same fixed Pi source has not been refitted.

Increasing NSIDE is an enabling part of the scientific test because it exposes smaller angular structures and permits more multipoles. In this resolution series the full-range residual RMS increases as the harmonic cutoff increases, while a common acoustic band improves from the coarsest grid and then stabilizes. High-ℓ transfer, residual foreground and split variability therefore matter increasingly to an accurate high-resolution assessment. The resolution dependence is therefore governed by the measured quantity, the harmonic interval and the effective observational response.

7.8 Seventy-gigahertz odd-even comparison

The first reported run uses the PR3 SEVEM foreground-subtracted 70 GHz temperature product at NSIDE 1024, over $\ell = 2$–2048. OE1 and OE2 constitute the primary observed coadd; the common intensity mask and official odd-even missing-pixel mask define the retained region. The retained fraction is 75.9100%, representing 9,551,690 retained pixels. The map output has observed RMS 84.9394 μK, SkyPi comparison RMS 81.2056 μK, a signed residual RMS **22.7240** μK and Pearson correlation **0.96358**. The final low-multipole ($\ell = 2$–29) residual harmonic RMS is 3.3223 μK. These values are the results of the reported SkyPi analysis; no additional processing of the original FITS pixels was performed for this manuscript.

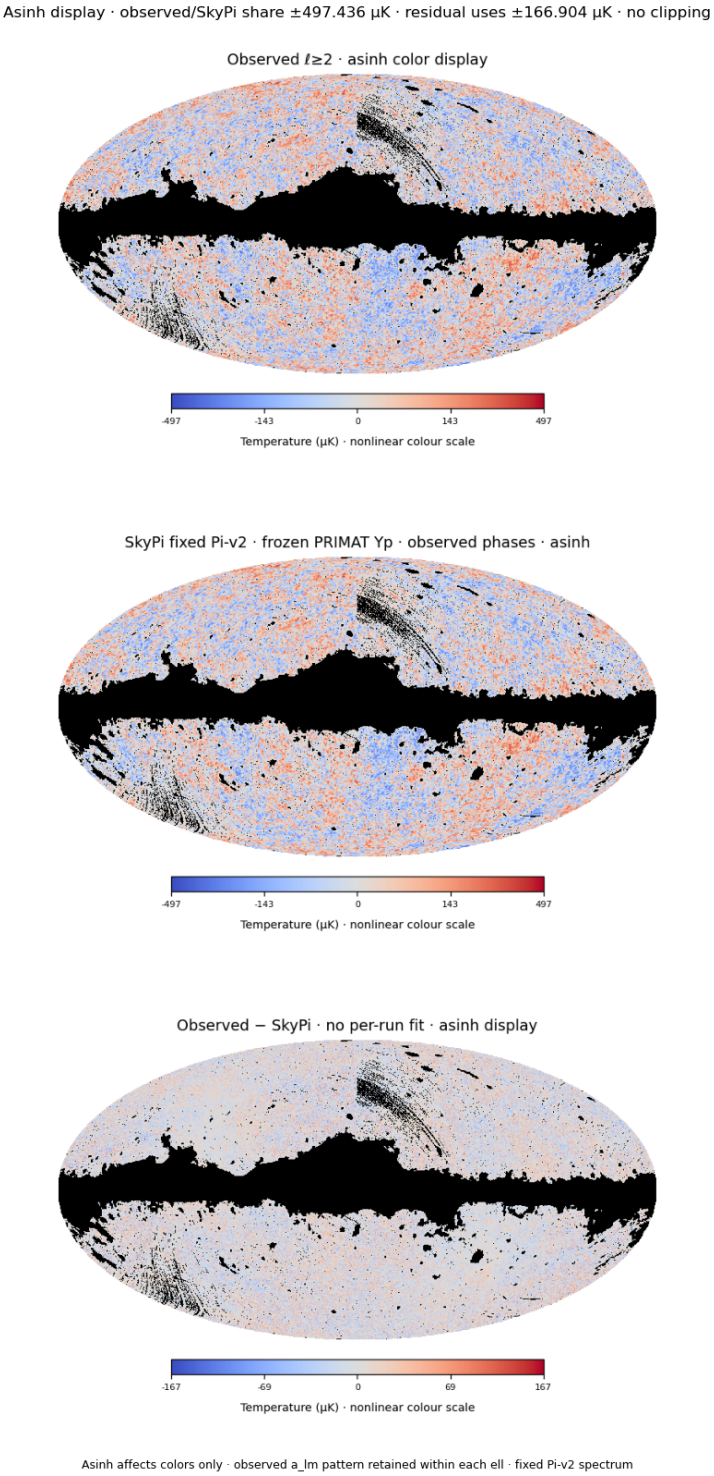


Figure 12: Saved 70 GHz SEVEM odd-even comparison figure from SkyPi. The panels display the recorded observed temperature, same-phase SkyPi comparison and signed residual using the run's asinh presentation. This is a visualization of the registered amplitude transformation applied to observed harmonic spatial structure; color and contrast choices are display operations, not changes to the source spectrum.

The odd-even cross-spectrum provides a second observable that is not interchangeable with the map residual. Table 12 gives the recorded differences in the OE-valid matched comparison. Across positive-power bands from $\ell = 30$ through 1499, the forward SkyPi band means are above the OE cross means by approximately 2–8%. The band 750–799 gives the largest reported difference in that interval, 8.06%. This demonstrates a broad numerical correspondence in the tested, positive cross-power bands under the declared operator; the percentages do not by themselves supply a covariance-based significance.

Multipoles	**OE cross mean D_ℓ (μK^2)**	**SkyPi forward mean (μK^2)**	**Difference**
2–29	361.098	342.871	−5.05%
30–199	1640.185	1673.329	+2.02%
200–449	1530.421	1592.193	+4.04%
450–549	661.516	691.444	+4.52%
550–649	465.297	489.821	+5.27%
650–749	287.664	302.723	+5.23%
750–799	250.947	271.164	+8.06%
800–999	131.129	138.986	+5.99%
1000–1499	14.671	15.159	+3.33%

Table 12: Band means generated by SkyPi for the 70 GHz OE cross-spectrum analysis. Differences are defined as SkyPi minus observed OE cross, divided by the positive OE cross mean. The low-ℓ band includes the frozen Commander-conditioned factors. High-ℓ negative or nearly vanishing cross-power bands are not used to claim a stable fractional precision.

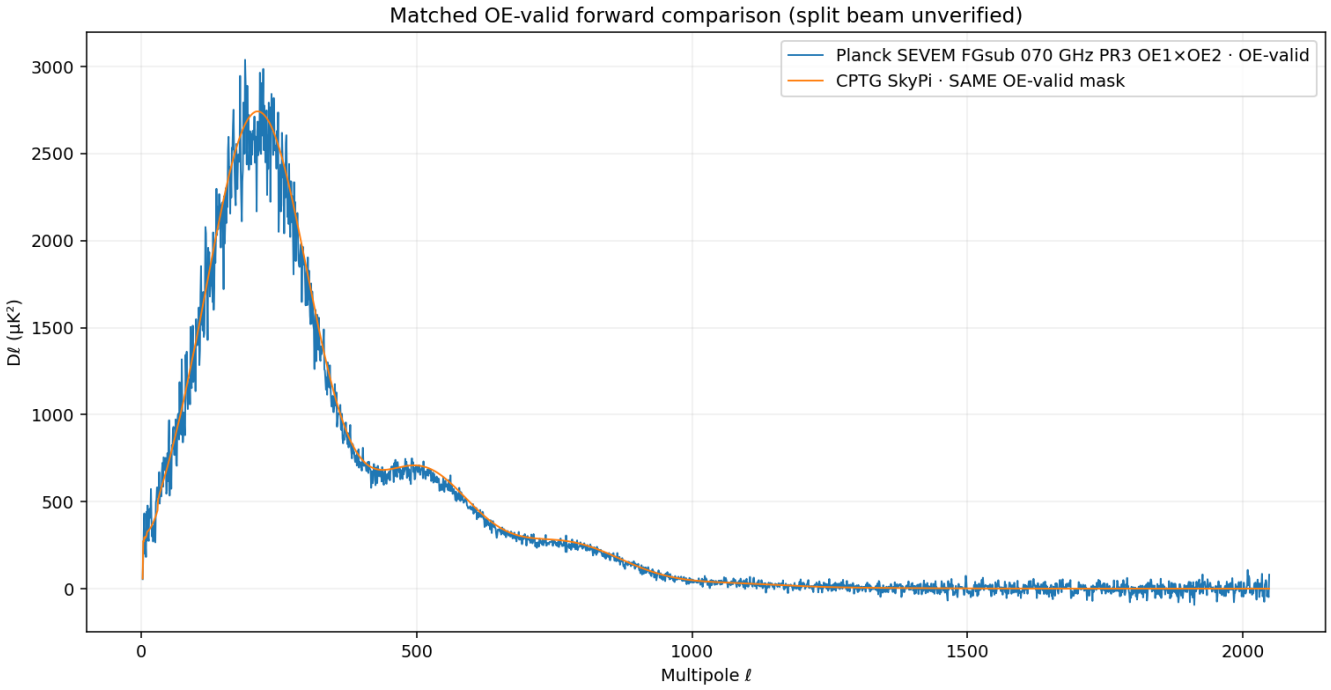


Figure 13: Planck PR3 SEVEM 70 GHz OE1×OE2 masked-sky cross-spectrum (blue) compared with the SkyPi fixed forward spectrum under the same OE-valid mask (orange). The plotted result is a direct visual counterpart of Table 12. The plot itself identifies the unverified split-beam response, so it should not be read as a calibrated likelihood fit.

7.9 High-multipole split-dependent power at 70 GHz

The same 70 GHz run permits an explicit examination of the rising high-multipole coadd auto-power. Over $1500 \leq \ell \leq 1999$, the mean coadd auto-D_ℓ is $794.832\,\mu\mathrm{K}^2$; the split half-difference mean is $795.727\,\mu\mathrm{K}^2$; and the OE cross mean is $-0.895\,\mu\mathrm{K}^2$. The SkyPi forward mean is $0.374\,\mu\mathrm{K}^2$. In the final 2000–2048 band, the corresponding values are 1061.835, 1057.662, 4.173 and $0.099\,\mu\mathrm{K}^2$. The near equality of coadd power and half-difference power follows the matched estimator identity and indicates that the large auto-spectrum in these bands is predominantly split-dependent rather than present at the same level in the OE cross signal.

Mean D_ℓ ($\mu\mathrm{K}^2$)	$\ell = 1500$–**1999**	$\ell = 2000$–**2048**
Observed coadd auto-power	794.832	1061.835
OE split half-difference	795.727	1057.662
OE1×OE2 cross-power	−0.895	4.173
SkyPi matched forward power	0.374	0.099

Table 13: High-multipole diagnostic terms from the recorded 70 GHz split analysis. Split-difference power is measured, but its attribution to instrumental noise alone is not certified; the effective split transfer and covariance are unverified.

A negative band-averaged cross spectrum is not a negative physical auto-power. It reflects fluctuations in the cross estimator about a small underlying shared signal under the specified mask and noise conditions. For this reason, the high-ℓ cross-to-forward ratio should not be interpreted as a high-precision percentage. The substantive result is the separation of a large coadd auto-spectrum from a much smaller split-shared cross component.

7.10 Two-hundred-and-seventeen-gigahertz full-mission comparison

The second case uses the PR3 SEVEM foreground-subtracted 217 GHz full-mission temperature map at NSIDE 2048, $\ell_{\max} = 4096$, with the Planck common intensity mask. The retained region contains 39,229,084 of 50,331,648 pixels, or 77.9412% of the analysis grid. The observed band-limited map RMS is $108.2765\,\mu\mathrm{K}$; the SkyPi amplitude-reweighted map RMS is $106.4233\,\mu\mathrm{K}$; the retained-region residual RMS is $\mathbf{13.4411}\,\mu\mathrm{K}$; and the Pearson correlation is **0.99231**. The $\ell = 2$–29 residual harmonic RMS is $4.2382\,\mu\mathrm{K}$.

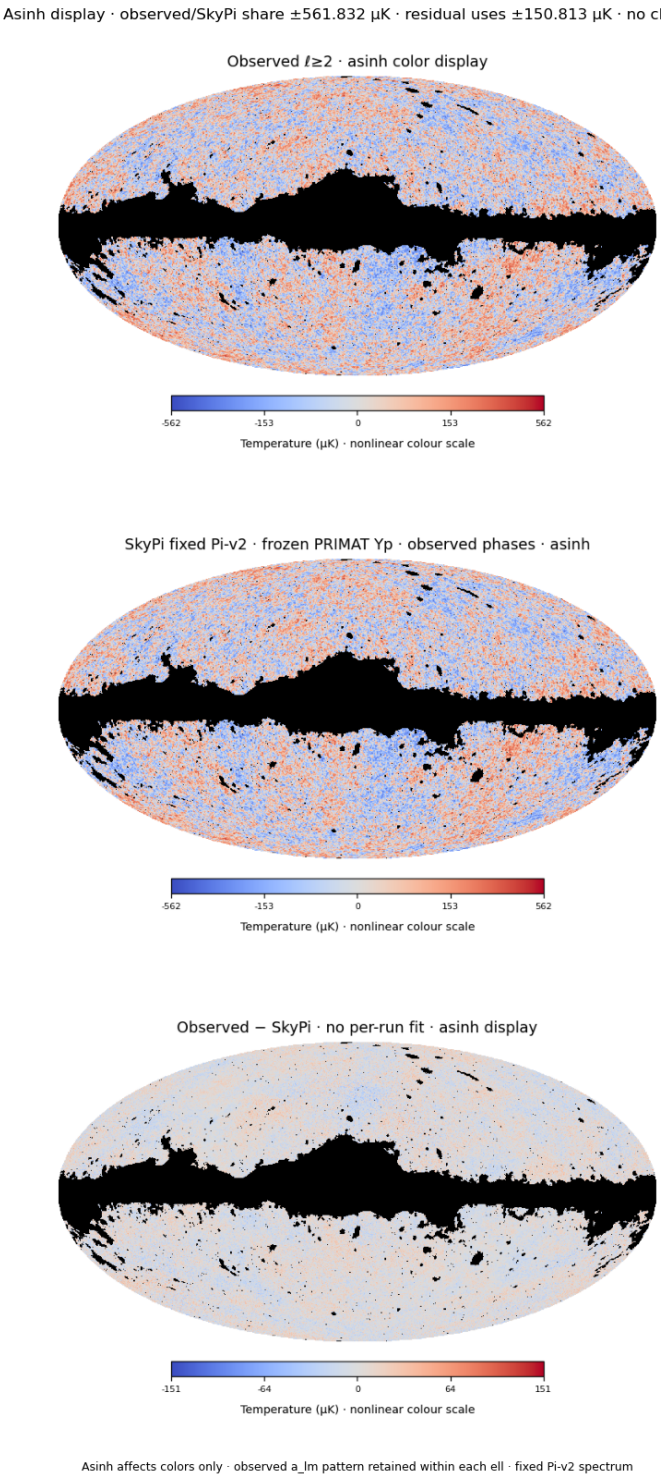


Figure 14: SkyPi-generated 217 GHz SEVEM full-mission observed, SkyPi and residual figure, using the saved asinh display palette. Observed harmonic spatial information is retained in the SkyPi map construction; the correlation measures map amplitude and morphology correspondence in that conditional representation. The masked residual is not an instrumental-noise-subtracted signal-only quantity.

In the direct full-mission masked pseudo-spectrum, the recorded SkyPi and observed power means are close in several intermediate bands. The 450–549 interval gives 1744.445 and 1750.567 μK^2, respectively, a difference of -0.35% when expressed as a ratio of the tabulated band means. In 550–649 the means are 1584.167 and 1581.120 μK^2 ($+0.19\%$); in 650–749 they are 1312.854 and 1314.825 μK^2 (-0.15%); and in 750–799 they are 1548.486 and 1542.875 μK^2 ($+0.36\%$). Over 1000–1499, however, the forward mean is 458.088 compared with 501.689 μK^2 observed, and at still higher multipoles the uncorrected observed auto-power exceeds the forward source substantially. These are *full-mission auto-spectrum* comparisons, not 217 GHz OE cross-spectrum measurements.

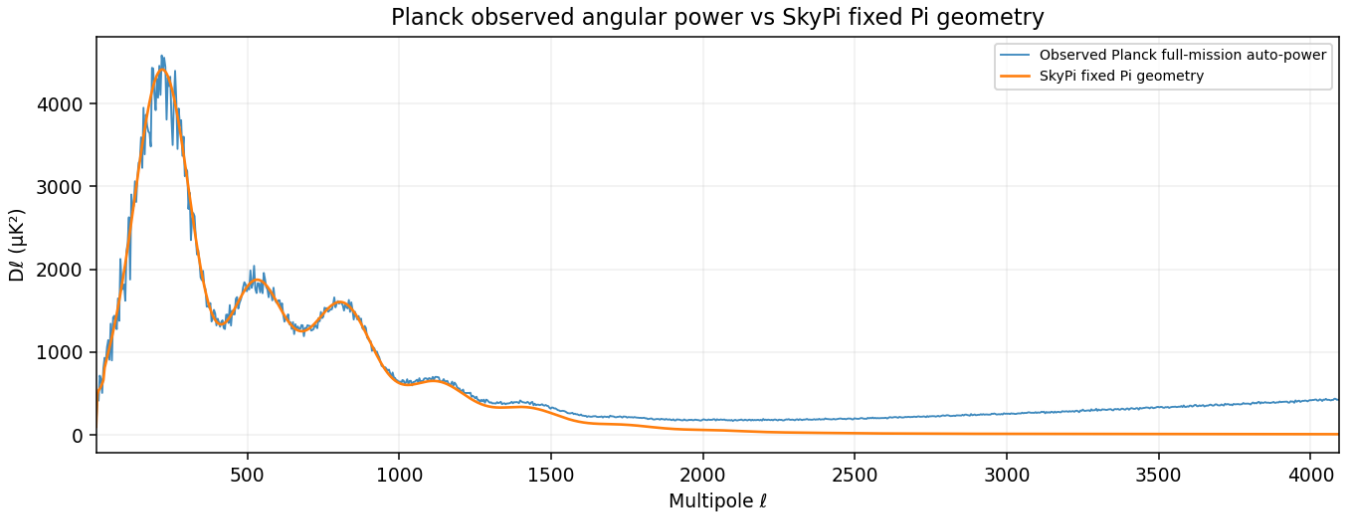


Figure 15: SkyPi masked-sky power-spectrum comparison for the 217 GHz full-mission SEVEM run. The displayed observed auto-spectrum contains split-unseparated noise, foreground residuals and measurement response. Its high-multipole rise must not be equated to a measured excess in a split cross-spectrum without a corresponding split analysis.

The 217 GHz masked-sky result is therefore a useful high-resolution demonstration of the workbench's range and operator visibility. It does not, by itself, identify whether the high-ℓ auto-power excess arises from noise, residual foreground, or mismatch in the approximate effective beam. An NPIPE 217 GHz beam FITS is not substituted into this PR3 cleaned-map run, and no revised 217 GHz pixel residual is reported here.

7.11 What the two runs establish together

The comparison can be summarized without merging their unlike measurement definitions. The 70 GHz result supplies a split-coadd map and an OE cross-spectrum with band differences measured under a shared window; the 217 GHz result supplies a full-mission map and a longer multipole-range auto-spectrum. Both calculations use the same frozen helium-updated Pi/CAMB source coordinates and registered low-ℓ factors, with their own declared observational responses. The observed map correlation and residual RMS are not directly comparable as a ranking because NSIDE, input sky, split policy, mask and nominal response differ.

Measured or recorded quantity	**70 GHz OE (F3)**	**217 GHz full (F4)**
Binary retained sky (%)	75.9100	77.9412
Mask	common∩OE	common only
Split / window	OE coadd; 1° C2	none; no apodization
Gaussian beam FWHM	13.08′	4.90′
Retained-region observed map RMS (μK)	84.9394	108.2765
Retained-region SkyPi map RMS (μK)	81.2056	106.4233
Observed minus SkyPi pixel RMS (μK)	**22.7240**	**13.4411**
Pearson map correlation	0.96358	0.99231
Final low-ℓ residual harmonic RMS (μK)	3.3223	4.2382
Phase-independent normalized amplitude distance	0.26971	0.12860

Table 14: Descriptive diagnostics and their immediate measurement settings (runs F3 and F4 in Table 15). Both mask classes use threshold 0.999; the 70 GHz OE mask is conservatively resampled to NSIDE 1024, while the 217 GHz full-mission common mask is native to NSIDE 2048. Both listed frequency-channel Gaussian responses are approximate cleaned-map proxies, not measured effective transfers.

8 Reproducibility in SkyPi

8.1 Fixed geometry and computational reproducibility

The reported analyses were performed with the CPTG SkyPi v3.9 scientific implementation [14]. The registered Pi-geometric source is unchanged from one Planck product or NSIDE setting to another. Repeating a comparison requires the stated observational input, fixed geometric configuration, angular resolution, multipole limit, mask and response settings. These conditions, rather than a new parameter fit, define each analysis.

8.2 Reproduce the reported results using the workbench

SkyPi calculates the maps, temperature distributions, harmonic tables and split spectra from the specified Planck observations. The input maps and masks are available through the Planck data distribution [7, 6]; the PR3 component-map filenames below follow the public Planck product naming. All temperature fields are read as thermodynamic `K_CMB` and converted to μK for map statistics. The registered CPTG source, including the frozen helium entry and low-ℓ comparison factors (Eqs. 6 and the source registry), is held fixed for each run.

Setting codes. **C** denotes `COM_Mask_CMB-common-Mask-Int_2048_R3.00.fits`, field `TMASK`; **O** denotes the official `COM_Mask_CMB-OE-Misspix-Mask-Int_2048_R3.00.fits`, field `TMASK`. For both, retained pixels satisfy $M \geq 0.999$ at the stated analysis resolution. **OE** means primary equal-weight OE1/OE2 coadd with a same-sky cross-spectrum and the *preserve* split-coupling policy. **W1** is a 1° spherical C2 quintic split-analysis window; **W0** denotes no split apodization. **I** is the named product's full-mission `INT_BEAM` (a proxy when applied to OE splits); **G**x is an explicitly assumed Gaussian FWHM of x arcmin. The displayed percentages are *binary* $f_{\rm sky}$, not f_{W^2}.

ID / observation	NSIDE / $\ell_{\rm max}$	Mask; retained sky	Response; estimator / window
P1 / Commander PR3 OE	2048 / 4096	C∩O; 76.3376%	I; OE, W1
P2 / SEVEM PR3 OE	2048 / 4096	C∩O; 76.3376%	G5; OE, W1
P3 / SMICA PR3 OE	2048 / 4096	C∩O; 76.3376%	I; OE, W1
P4 / NILC PR3 OE	2048 / 4096	C∩O; 76.3376%	I; OE, W1
P5 / SMICA-NOSZ PR3 OE	2048 / 4096	C∩O; 76.3376%	G5; OE, W1
F1 / SEVEM FGsub 100 GHz OE	2048 / 4096	C∩O; 76.3376%	G9.66; OE, W1
F2 / SEVEM FGsub 217 GHz OE	2048 / 4096	C∩O; 76.3376%	G4.90; OE, W1
F3 / SEVEM FGsub 70 GHz OE	1024 / 2048	C∩O; 75.9100%	G13.08; OE, W1
F4 / SEVEM FGsub 217 GHz full	2048 / 4096	C; 77.9412%	G4.90; full, W0
C1 / Commander PR3 full	1024 / 3071	C; 77.7964%	I; full, W0
C2 / Commander PR3 full	2048 / 4096	C; 77.9412%	I; full, W0
N1 / Commander PR4 full	256 / 512	C; 76.6440%	G5; full, W0
N2 / Commander PR4 full	512 / 1024	C; 77.4641%	G5; full, W0
N3 / Commander PR4 full	1024 / 2048	C; 77.7964%	G5; full, W0
N4 / Commander PR4 full	4096 / 8192	C; 77.9412%	G5; full, W0
H1 / HFI R4 100 GHz full	2048 / 4096	C; 77.9412%	G9.66; full, W0
H2 / HFI R4 217 GHz full	2048 / 4096	C; 77.9412%	G4.90; full, W0
D1 / HFI PR3 545 GHz dust	2048 / 4096	none; 100%	observed brightness only; no CMB response

Table 15: Run-specific measurement configurations for the reported tables and figures. Masks, split windows, Gaussian or FITS-response assumptions and retained sky fractions form part of each numerical result. A full-mission response used with OE splits is a proxy, not a certified split transfer.

8.3 Exact temperature inputs and split files

The PR3 component-separated tests select the following full-mission files for header and response information. For the four OE rows, the primary observed temperature and measured split spectrum come from the matching `oe1` and `oe2` products, not from full-mission temperature pixels. All three files in a product family have the native temperature field `I_STOKES`.

- **Commander PR3:** `COM_CMB_IQU-commander_2048_R3.00_full.fits`; split maps `COM_CMB_IQU-commander_2048_R3.00_oe1.fits` and `COM_CMB_IQU-commander_2048_R3.00_oe2.fits`.
- **SEVEM PR3:** `COM_CMB_IQU-sevem_2048_R3.00_full.fits`; split maps `COM_CMB_IQU-sevem_2048_R3.00_oe1.fits` and `COM_CMB_IQU-sevem_2048_R3.00_oe2.fits`.
- **SMICA PR3:** `COM_CMB_IQU-smica_2048_R3.00_full.fits`; split maps `COM_CMB_IQU-smica_2048_R3.00_oe1.fits` and `COM_CMB_IQU-smica_2048_R3.00_oe2.fits`.
- **NILC PR3:** `COM_CMB_IQU-nilc_2048_R3.00_full.fits`; split maps `COM_CMB_IQU-nilc_2048_R3.00_oe1.fits` and `COM_CMB_IQU-nilc_2048_R3.00_oe2.fits`.
- **SMICA-NOSZ PR3:** `COM_CMB_IQU-smica-nosz_2048_R3.00_full.fits`; OE1 `COM_CMB_IQU-smica-nosz_2048_R3.00_oe1.fits`; OE2 `COM_CMB_IQU-smica-nosz_2048_R3.00_oe2.fits`. Official OE missing-pixel mask.
- **SEVEM foreground-subtracted 100 GHz (F1):** `COM_CMB_IQU-100-fgsub-sevem_2048_R3.00_full.fits`; OE1 `COM_CMB_IQU-100-fgsub-sevem_2048_R3.00_oe1.fits`; OE2 `COM_CMB_IQU-100-fgsub-sevem_2048_R3.00_oe2.fits`.
- **SEVEM foreground-subtracted 217 GHz OE (F2):** `COM_CMB_IQU-217-fgsub-sevem_2048_R3.00_full.fits`; OE1 `COM_CMB_IQU-217-fgsub-sevem_2048_R3.00_oe1.fits`; OE2 `COM_CMB_IQU-217-fgsub-sevem_2048_R3.00_oe2.fits`. Distinct from the full-mission analysis below.
- **HFI PR4 100 and 217 GHz full-frequency tests:** `HFI_SkyMap_100_2048_R4.00_full.fits` and `HFI_SkyMap_217_2048_R4.00_full.fits`; NSIDE 2048, $\ell_{\max} = 4096$, common intensity mask.
- **HFI 545 GHz observational dust mode:** `HFI_SkyMap_545_2048_R3.01_full.fits`; NSIDE 2048 and $\ell_{\max} = 4096$, observed surface brightness, not a CMB comparison.
- **Commander PR4 NSIDE sequence:** the same `COM_CMB_IQU-commander_4096_R4.00_full.fits` for all four runs; temperature field `TEMPERATURE`, units `K_CMB`; no split files.
- **SEVEM foreground-subtracted 70 GHz:** `COM_CMB_IQU-070-fgsub-sevem_1024_R3.00_full.fits`, with `COM_CMB_IQU-070-fgsub-sevem_1024_R3.00_oe1.fits` and `COM_CMB_IQU-070-fgsub-sevem_1024_R3.00_oe2.fits` as the primary split maps. Field `I_STOKES`.
- **SEVEM foreground-subtracted 217 GHz:** `COM_CMB_IQU-217-fgsub-sevem_2048_R3.00_full.fits`, field `I_STOKES`; full-mission map only.

8.4 Mask, apodization and response selections

The numerical settings in Table 15 were read from each run's recorded configuration. The common mask **C** is taken from the official NSIDE-2048 Planck PR3 intensity mask, `TMASK` field, with binary threshold $M \geq 0.999$. The official OE missing-pixel mask **O** is combined by binary intersection *before* the one-degree C2 quintic split-analysis taper **W1**. At NSIDE 2048 the intersection retains 38,421,990 of 50,331,648 pixels (76.3376%); the full-sky mean of the *squared* split-analysis weight is approximately 0.531979. The weighted harmonic mask therefore cannot be identified with the binary retained-pixel percentage. Full-mission cases use **W0**, no split-apodization taper. The exact percentage retained for each case appears beside the corresponding measurement in the Results section and in the full settings table.

When an NSIDE-2048 mask is transferred to an NSIDE-1024, 512 or 256 analysis grid, SkyPi uses conservative mean downgrade and an eligibility threshold of 0.999. Consequently, the Commander PR4 NSIDE sequence does *not* have exactly the same binary sky fraction at each resolution: N1–N4 retain 76.6440, 77.4641, 77.7964 and 77.9412%, respectively. For NSIDE 4096, the NSIDE-2048 parent flags are replicated onto the finer grid and supply no additional native mask information. This mask sampling must be distinguished from the independent resolution of the temperature map itself. The observed-only 545 GHz dust case uses no CMB foreground mask.

For P1, P3, P4, C1 and C2 the temperature response is read from the corresponding full-mission `INT_BEAM` data. For P2 and P5 the recorded choice is the nominal 5-arcmin component response. F1, F2, F3, F4, H1 and H2 use their stated

nominal frequency-channel Gaussian widths. The Commander PR4 N1–N4 sequence uses the 5-arcmin published nominal smoothing. No separate pixel-window correction was applied in these recorded configurations. The response applied to a component's OE comparison is a declared full-mission or nominal proxy, not a separately certified OE split transfer. The OE *preserve* policy does not substitute an inverse-coupling solution for the missing effective split transfer.

For OE runs, the selected OE1 and OE2 maps, not the full-mission temperature pixels, provide the observed coadd and cross-spectrum. The full-mission product supplies its family identification, metadata and applicable response reference. For full-mission entries no OE/HM split was used. The fixed CPTG source, the observed-phase map-construction convention, and the recorded residual definitions remain as defined above. The analysis-mask sampling-convergence diagnostic was not requested for the runs tabulated here; it would constitute a separate operator validation rather than a change to CPTG geometry.

To reproduce a tabulated value, select the stated temperature product and field, the run ID's NSIDE/$\ell_{\max}$, the **C** and/or **O** masks as required, the threshold and downgrade/replication rule, and the listed response and split/window policy. Then run *Check Inputs* and generate the SkyPi comparison. The residual map, Temperature Statistics and spectrum diagnostics must be compared with the corresponding estimator in the same row. A result obtained with a different mask, sky fraction, apodization, split selection, or harmonic support is a distinct measurement even if the fixed geometric source is identical.

9 Discussion

SkyPi's principal scientific role is to implement and interrogate the fixed CPTG geometric CMB solution in actual Planck observation coordinates. This interpretation follows directly from the hierarchy established in the CPTG unified framework, its canonical Pi-branch register, and the original CMB map-closure paper [4, 1, 3]. The geometric source is not recalibrated for a new frequency product or changed NSIDE. Instead, each observation is processed through a declared response, mask, split estimator and harmonic support. The software thereby replaces repeated cosmological-row fitting *within the registered comparison protocol* with a fixed geometric input, while keeping measurement calibration distinct from geometric freedom.

The accumulated evidence spans multiple observational families rather than one selected component-separation algorithm. The original NSIDE-256 study established map-field closure and tested generic null envelopes. At NSIDE 2048 the frozen source under a common OE mask yields high map correspondence across four directly matched Planck component pipelines; the OE cross-spectrum band means follow a closely grouped acoustic pattern. Separately tested SMICA-NOSZ and foreground-subtracted frequency products broaden the observational base while retaining their own declared operators. The fixed-source Commander PR4 resolution series demonstrates that the resolution of the observation grid changes the result in a measurable way: the common 450–511 band improves appreciably from NSIDE 256 to 512, then stabilizes across still-finer grids. The 70 and 100 GHz OE, 217 GHz OE and full-mission cases expose complementary frequency and harmonic-range behavior. Raw HFI and 545 GHz dust tests extend the workbench's observational coverage without being represented as foreground-cleaned CMB matches.

NSIDE is therefore not a mere visualization setting. More pixels and modes can reveal correspondence obscured at coarse sampling; they also make additional small-scale residuals and response limitations measurable. The Commander PR4 common-band comparison improves between NSIDE 256 and 512 and remains nearly constant at higher resolutions (Table 11). Full-range map residuals include successively larger multipole intervals and therefore measure a different quantity at each resolution. Together, the fixed-band and full-range analyses characterize the resolution dependence of the observed geometric correspondence.

The phase-locked map transformation and the independent observed masked-sky spectrum are also different observables. The high map correlations demonstrate close amplitude-weighted correspondence within an observed spatial scaffold. The split OE cross-spectra provide additional amplitude/power tests for the same frozen source without relying on the spatial coincidence of individual peaks. Neither statistic should be silently converted into the other. The HM/OE half-difference decomposition further shows why high-ℓ auto-power may be dominated by split-variable contributions; its physical attribution requires the appropriate split noise model and response.

Published Planck PR4 work confirms the importance of frequency-specific QuickPol transfer functions, detector-set construction, sky selection and noise covariances [9]. Their likelihood and fitted nuisance framework is a distinct statistical object from SkyPi's present source-locked, masked-forward spectrum. Likewise, the earlier Version II CPTG Pi-branch reference documents fixed-row Plik, NPIPE CamSpec and SPT/ACT profile controls in its own stated likelihood protocols [2]; the current manuscript does not re-label those earlier analyses as new SkyPi results or claim that the present map

work has replaced full Planck likelihood estimation.

Taken together, the evidence provides a coherent progression from the original geometric map closure to higher-resolution, operator-aware real-sky testing. More precision will come from matched effective cleaned-map/split beams, independently characterized covariance, and uniform-band NSIDE sweeps. Those are measurement improvements, not a reason to alter the CPTG fixed Pi relations.

10 Conclusion

CPTG SkyPi is the computational endpoint of the fixed-geometric CMB comparison-map research program. Its operating principle is a registered CPTG Pi-based comparison row in place of a per-dataset adjustable cosmological-row fit, carried through explicit observational coordinate and measurement operators. The original CMB map paper demonstrated field-coordinate closure and finite null-envelope discrimination at NSIDE 256. SkyPi expands that construction into a local high-resolution scientific workbench with measured Planck spectra, same-phase map reconstruction, HM/OE decomposition, reproducible provenance and optional NSIDE-4096 analyses.

Among the broader collection of completed Planck-family tests, the most directly comparable cross-pipeline subset uses NSIDE 2048, $\ell_{\max} = 4096$, and a common OE-valid mask. Commander, SEVEM, SMICA and NILC then give map correlations 0.993454–0.995103 and retained-region residual RMS values 8.8105–10.2111 μKwithout changing the registered fixed Pi spectrum. Their acoustic OE cross-band comparisons provide a separate harmonic-power view of that correspondence. In the same-source Commander PR4 resolution series, the $\ell = 450$–511 band-mean discrepancy improves from about 2.57% at NSIDE 256 to about 1.10% at NSIDE 512 and remains near that level at 1024 and 4096. A higher all-mode RMS at NSIDE 4096 reflects, in part, the much larger harmonic interval being included, and is not evidence that the matched lower multipoles lost the same level of agreement.

Across the matched four NSIDE-2048 PR3 OE products, and in the additional SMICA-NOSZ and frequency-separated analyses, the temperature-statistics distributions are principal visual and quantitative results: the fixed-source comparisons closely reproduce the observed retained-region temperature populations, while the matched residuals remain much narrower than the physical temperature span. Their evidential role is distinct from, and complementary to, map correlation and angular-power consistency.

SkyPi thus supplies a concrete, reproducible setting in which a fixed CPTG geometric CMB construction can be followed from its source registry to the angular structure of observed skies at progressively finer resolution. Its decisive contribution is the integration of geometric rigidity, observable-level distinctions and recorded numerical results across Planck products, rather than any one image or software feature.

Code and Data Availability

The CPTG SkyPi scientific software workbench is archived on Zenodo [14]; its research source lineage is also identified at `https://github.com/CLG2025/CPTG` [15]. The run-by-run analysis configurations, including mask threshold, actual retained sky, mask resampling, apodized window, beam response and split estimator, are specified in Table 15 and Section 8.4. The observational inputs are standard Planck temperature and mask FITS products; these should be obtained from their official data distributors. The paper supplies the numerical summaries and figures needed to examine its findings. Readers can regenerate the underlying maps, spectra, temperature-statistics plots and output metadata directly in SkyPi using the product choices and analysis settings in Table 15. The paper is self-contained in its results, and the corresponding calculations can be repeated within SkyPi.

The displayed figures were generated from SkyPi outputs, or, for the NSIDE plots, from the numerical values printed in the manuscript. The figures illustrate the reported analyses; they do not substitute for running the corresponding Planck observations through SkyPi. Matching input maps, mask conventions, response choices and source identity remain necessary to reproduce a numerical result, and the reported agreement is not a covariance-calibrated Planck likelihood significance.

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